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RunConnect: An Interactive Web Platform to Engage Runners and Streamline Marathon Registration

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CHAPTER I

INTRODUCTION

A marathon is a long-distance road race that continues to gain popularity worldwide as more individuals pursue healthier lifestyles and community-based fitness activities [1]. Organizing a marathon event requires collaboration among various stakeholders, including organizers, healthcare officers, safety officers, support teams, and marathoners, to ensure a smooth, safe, and engaging experience for all participants [2]. As stated in [3], successful marathon events depend on the effective coordination of essential services such as communication, management, support operations, entertainment, health, and safety. Although digital technologies such as registration websites, social media platforms, and mobile applications linked to wearable devices have been utilized to support specific aspects of marathon activities in recent years, these tools remain limited, fragmented, and unable to meet the increasing demands of modern marathon events. Crucially, many existing platforms focus solely on registration logistics, often overlooking the vital aspects of participant safety monitoring and the verification of runner capability during the race. As participation grows, the need for a more intelligent, unified, and safety-oriented system becomes critical.

However, despite the increasing popularity of marathon events, existing registration systems continue to experience recurring issues that negatively affect operational efficiency, safety standards, and user satisfaction. Studies indicate that participants in large-scale marathon events frequently report low satisfaction with registration platforms due to slow system performance, limited payment options, and frequent data-handling errors [4]. A significant gap in current systems is the reliance on subjective self-assessment, where runners manually select their category without verification. This lack of data validation often leads to participants joining race categories beyond their physical capability, increasing the risk of injury and logistics mismanagement. Furthermore, runners increasingly expect digital platforms to offer personalized recommendations and seamless engagement; however, most current systems lack the capability to tailor events based on individual preferences and verified

participation history [5]. Additionally, the lack of real-time visibility during the race leaves organizers with "blind spots" regarding runner locations, complicating safety response and medical deployment. Research suggests that event management systems that fail to integrate hybrid algorithms, performance analytics, or automated matching techniques will struggle to maintain fairness, accuracy, and scalability [6]. These findings highlight a significant gap in existing marathon registration platforms, which remain fragmented, error-prone, and insufficient in delivering personalized, data-driven services. Without the adoption of advanced technological solutions such as pace verification engines and optimized slot allocation mechanisms, both runners and organizers will continue to face challenges including duplicated registrations, unfair slot distribution, safety risks, and limited accessibility.

In response to the identified challenges and research gaps, this study aims to design and develop an intelligent web-based marathon management system that streamlines the registration process, enhances user interaction, improves race-day safety, and supports organizers in managing large-scale marathon events more efficiently. The proposed system will incorporate a Performance Verification Module to calculate runner pace from historical data, a hybrid recommendation algorithm to deliver personalized and capability-matched event suggestions, and a weighted matching algorithm to ensure transparent and equitable slot allocation [5], [6]. Furthermore, the system will integrate a Checkpoint Analytics feature to provide organizers with real-time tracking of participant progress. By integrating these advanced analytical features, the study seeks to provide a more reliable, accessible, and user-centered digital solution that benefits both runners and event organizers.

Theoretical Framework

This study is anchored on theories related to marathon race management, which offer a structured understanding of how running events are organized and how participants experience them. According to [7], marathon management involves five core areas communication, service management, environment and entertainment, medical services, and post-race engagement which collectively shape the technical and

emotional satisfaction of runners. These concepts are further supported by World Athletics guidelines [3] and studies highlighting the role of event image, service quality, and athlete engagement in shaping runner experiences and sports tourism outcomes [8], [9], [10], [11]. These theoretical foundations justify the need for a digital system designed to enhance communication, improve management processes, and support safer and more engaging marathon events.

The study is also grounded in algorithmic theories, particularly Hybrid Recommendation Algorithms and Weighted Matching Algorithms. Hybrid models, which combine content-based and collaborative filtering, are shown to produce more accurate and stable recommendations by utilizing both user behavior and item characteristics, as supported by [12]. Weighted matching, meanwhile, operates on the principle of optimizing decisions through quantitative evaluation of multiple criteria, as established in [13]. These algorithmic theories justify the system's capability to deliver personalized marathon recommendations and to allocate event slots fairly based on user profiles, preferences, and priority factors.

DeLone and McLean Information Systems Success Model To measure the effectiveness of the developed software, the study uses the DeLone and McLean IS Success Model. This model states that the success of an information system relies on three quality factors: System Quality, Information Quality, and Service Quality [14].

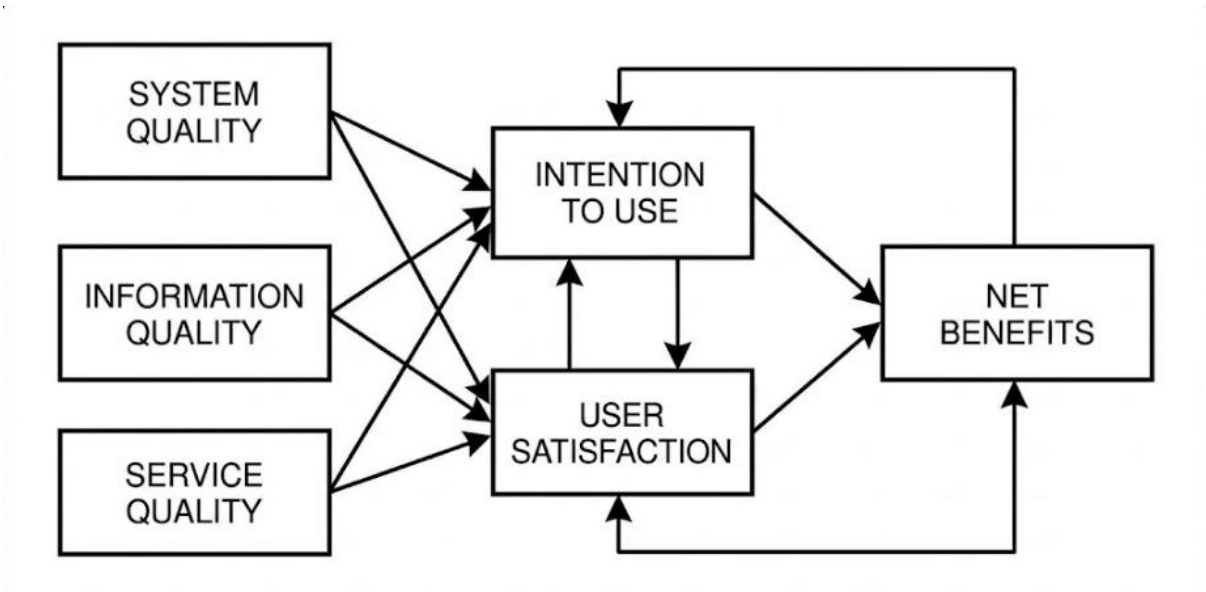


Figure 1. DeLone and McLean IS Success Model [1]

Figure 1 shows the relationship between system qualities and user satisfaction. System Quality refers to the technical performance, meaning the website must load fast and be error-free. Information Quality refers to the accuracy of event details and recommendations. Service Quality refers to the support provided to users. The researchers used this framework to ensure that *RunConnect* meets these quality standards, which is expected to lead to higher user satisfaction and better slot allocation outcomes.

Technology Acceptance Model (TAM) The researchers also considered user adoption using the Technology Acceptance Model (TAM). Introduced by Davis, TAM suggests that users will accept a new system if they perceive it as Useful and Easy to Use [15].

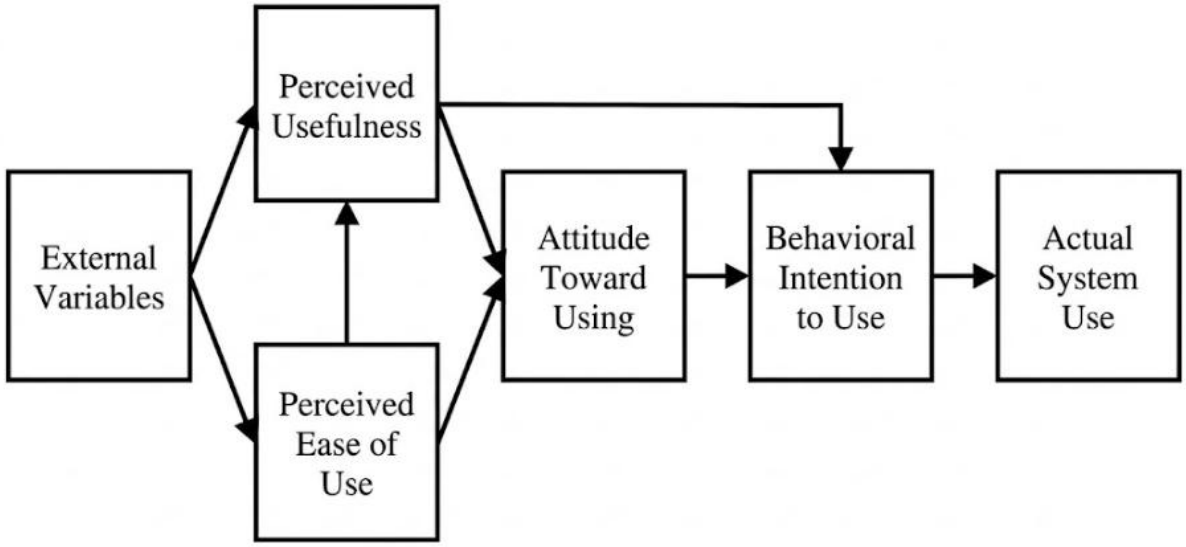


Figure 2. Technology Acceptance Model (TAM) [2]

Figure 2 illustrates the factors influencing a user's decision to use a system. Perceived Usefulness refers to the runner's belief that the system helps them secure

slots and find events more efficiently than manual methods. Perceived Ease of Use refers to the interface design, ensuring that the registration process is simple. This theory guided the researchers in designing a user-friendly interface to encourage the local running community to shift from manual to digital registration.

User Engagement Theory Beyond simple acceptance, the study considers how to keep users interested using User Engagement Theory. O'Brien and Toms define engagement as a quality of user experience characterized by Aesthetic Appeal, Perceived Usability, and Endurability [16].

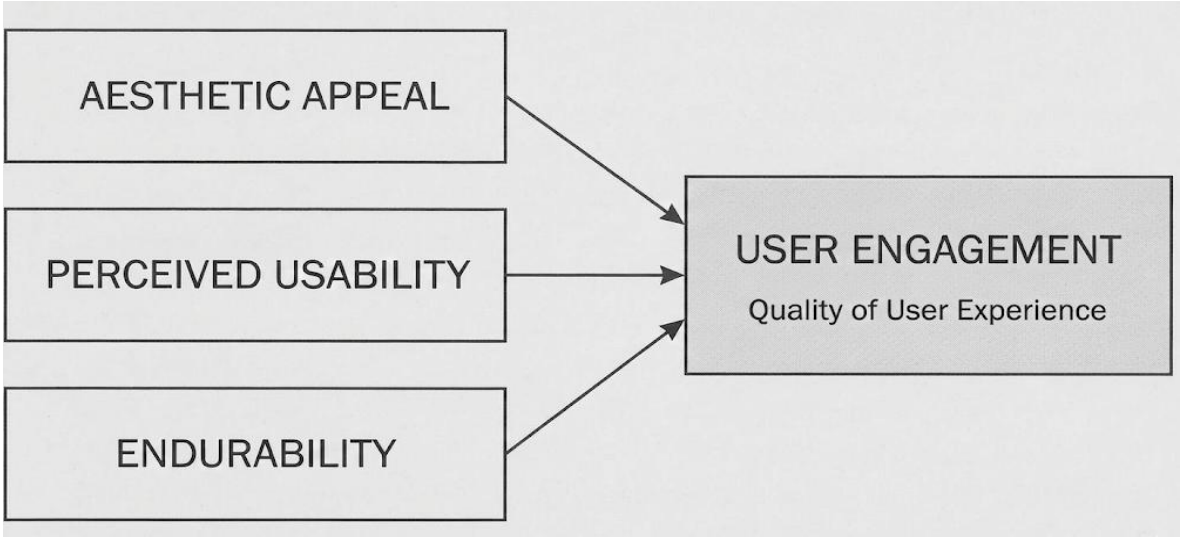


Figure 3. User Engagement [3]

Figure 3 presents the components of user engagement. It highlights that a system must look good (Aesthetic Appeal) and be easy to navigate (Usability) to encourage users to return (Endurability). This theory is relevant to *RunConnect* because the platform is not just a one-time registration form; it includes interactive features like the dashboard and personalized recommendations. These features are designed to keep runners engaged and returning to the platform for future races

Decision Support System (DSS) and Algorithmic Theory Since *RunConnect* provides event suggestions and fair slot allocation, it functions as a Decision Support System (DSS). Sprague and Carlson define DSS as systems that use data to help

users make decisions [17]. This is implemented using Hybrid Recommendation Theory [12] and Weighted Matching Theory [13].

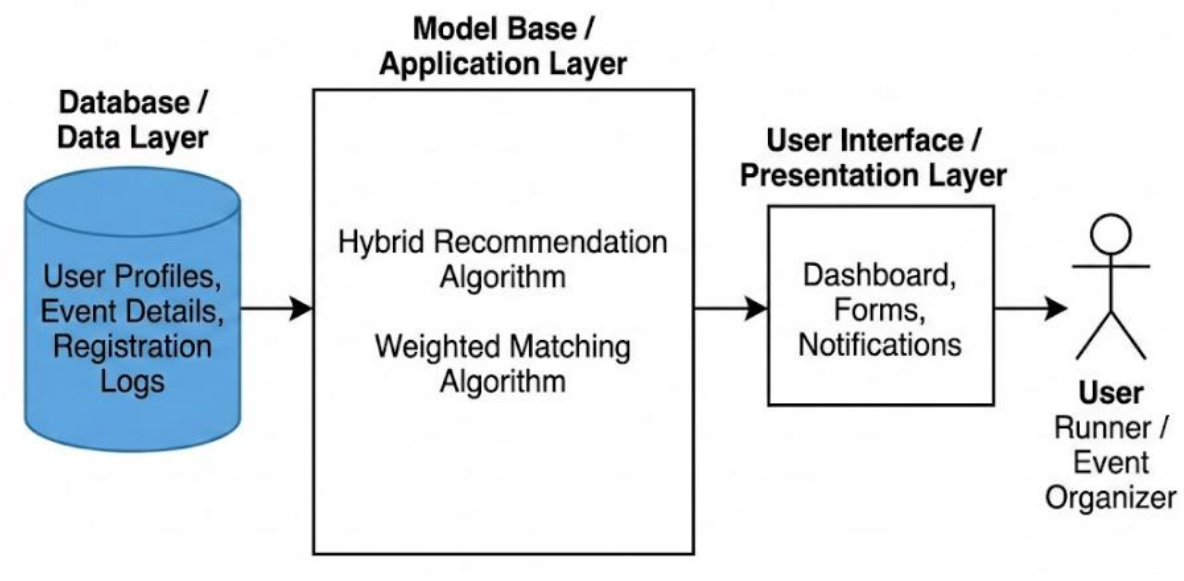


Figure 4. Decision Support System Architecture [4]

Figure 4 presents the Decision Support framework of the system. It shows how data (user profiles and event details) is processed by the algorithms to produce outputs. The Hybrid Recommendation Algorithm supports the runner's decision by suggesting suitable events, while the Weighted Matching Algorithm supports the organizer's decision by automating fair slot distribution. This structure ensures that the system provides logical and data-driven solutions for both user groups.

Finally, the System Development Life Cycle (SDLC) provides the methodological basis for building the platform in a structured and iterative manner. The Iterative SDLC Model is particularly suitable because it allows continuous refinement of system features, aligning development with user feedback and technical requirements. This framework ensures that key components such as recommendation algorithms, user interfaces, and database functions are systematically designed and tested. The application of SDLC principles supports the creation of a reliable, user-centered system that enhances marathon event management and overall runner experience.

Conceptual Framework

The conceptual framework of *RunConnect* serves as the blueprint that illustrates how the theories, algorithms, and processes are interconnected to achieve the main objective of the study to develop an interactive web platform that enhances user engagement while simplifying marathon registration and management.

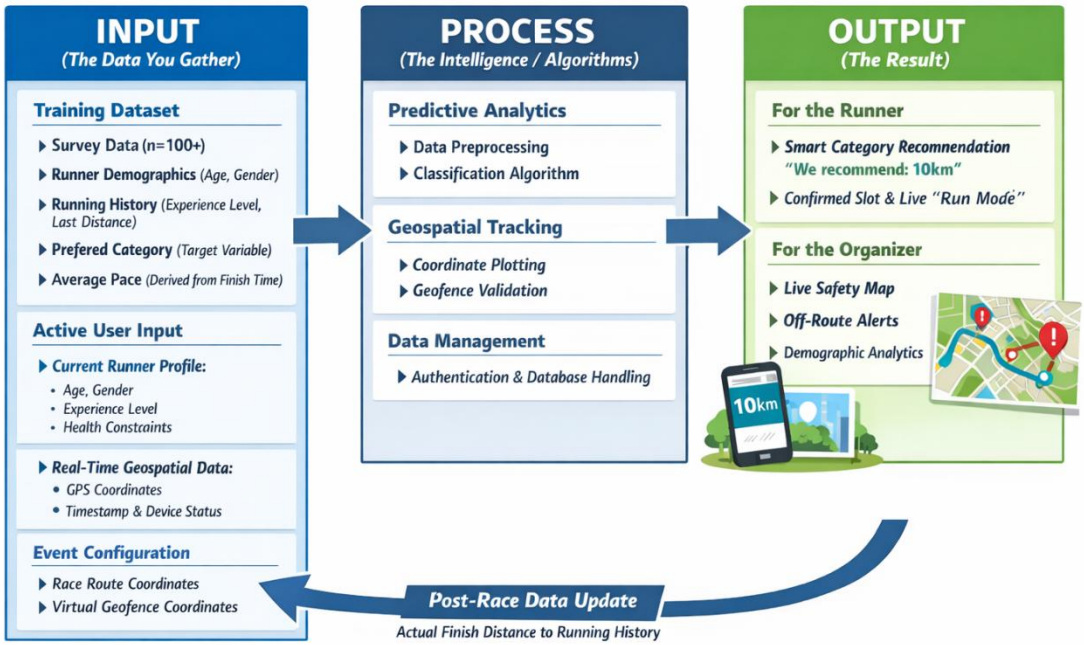


Figure 5. Conceptual Framework [5]

As illustrated in Figure 5, the system operates on an Input-Process-Output model that integrates machine learning and geospatial tracking. The input layer gathers a Training Dataset derived from runner surveys to establish demographic performance trends, alongside Active User Inputs which include real-time GPS coordinates and specific health constraints. These inputs feed into the processing layer, where a Predictive Analytics Module uses a classification algorithm to match user profiles against historical patterns for accurate categorization, while a Geospatial Tracking

Engine simultaneously maps live GPS streams to validate runner locations against geofenced race routes.

The system produces actionable outputs that enhance both decision-making and safety. Runners receive Smart Category Recommendations that are scientifically matched to their profile and health data, while organizers are equipped with a Live Safety Map that provides real-time visualization of participants and immediate alerts for off-route deviations. Completing the cycle, a Post-Race Feedback Loop captures actual finish data and feeds it back into the system, dynamically updating the runner's history and refining the predictive model's accuracy for future events.

Statement of the Problem

Despite the growing adoption of digital platforms for marathon registration, existing systems still suffer from limited automation, weak personalization, and inefficient slot allocation. Most platforms merely collect registration data and process payments, but they fail to utilize intelligent algorithms that could enhance fairness, accuracy, and user engagement. Because of this, several functional and methodological issues remain unresolved, prompting the need for a system that integrates computational intelligence into the marathon registration process.

- How will the hybrid recommendation algorithm accurately identify and suggest marathon events that match the preferences, running history, and location of different users?

- How will the weighted matching algorithm ensure fair, efficient, and conflict-free allocation of marathon slots, especially during peak registration periods?

- How will the web platform address existing issues in marathon registration such as slow processing, duplicate entries, limited personalization, and lack of real-time updates?

- How will the system maintain performance, scalability, and reliability when handling increasing numbers of users, events, and simultaneous registration requests?

General Objective

The general objective of this study is to develop RunConnect, an intelligent web platform that integrates machine learning for runner categorization and geospatial tracking for real-time event safety.

Specific Objectives

1. To collect runner demographics and self-reported running history to serve as the foundation for automated profiling and category recommendations.
2. To design a runner–event recommendation and matching model using hybrid recommendation and weighted matching algorithms for marathon registration.
3. To design a Real-Time Geospatial Tracking Module that visualizes runner locations on an interactive map and triggers Geofence Alerts for off-route participants.
4. To evaluate the performance, usability, and accuracy of the developed system, including the hybrid recommendation and weighted matching algorithms, using Precision, Recall, and F1-Score for algorithmic accuracy, and Weighted Mean based on a 5-point Likert Scale for user satisfaction and usability.

Scope and Limitation of the Study

The study focuses on the development of *RunConnect*, an intelligent web-based platform with mobile-responsive features. The system's core intelligence relies on a Training Dataset gathered from local runners; therefore, the accuracy of the Predictive Analytics Module is limited to the patterns identified within this specific dataset. For the

Geospatial Tracking, the system utilizes Browser-Based GPS. Tracking is active only when the runner keeps the web application open in "Run Mode" on their smartphone with an active internet connection (Mobile Data). The system does not function offline and depends on the user's device battery and signal strength. The study does not cover native mobile app development (APK/IPA). Furthermore, the Health Constraint feature is self-reported by the user and serves as a safety filter in the Weighted Matching Algorithm, not as a medical diagnosis tool.

Significance of the Study

This study will address long-standing challenges in marathon event registration by introducing automation, personalization, and fair slot allocation. Consequently, the implementation of this system will directly benefit the following groups:

1. **Runners / Participants**

   Runners will receive personalized event recommendations based on their preferences and performance data, ensuring better event alignment. They will also experience faster registration and fairer slot allocation through the weighted matching algorithm.

2. **Event Organizers**

   Organizers will benefit from an efficient registration system that reduces errors, removes duplicate entries, and prevents overbooking. They will also gain analytics that support data-driven planning and real-time participant management.

3. **Running Community / Clubs**

   Running communities will enjoy a more organized and inclusive participation system. Personalized event suggestions will increase engagement

and encourage more runners to join local and regional events.

4. **Local Government Units (LGUs)**

LGUs will improve event coordination using accurate registration and tracking data. This will enhance safety planning, crowd management, and tourism-related decision-making.

Definition of Terms

To better understand the concepts used in this study, the following terms are defined based on how they are applied within the system *RunConnect*.

Algorithm Parameters – These are predefined values and conditions used by the system's algorithms (Hybrid Recommendation and Weighted Matching) to process data and generate accurate recommendations or slot allocations.

Content-Based Filtering – A recommendation technique that suggests marathon events based on a runner's personal attributes, preferences, and past activities.

Collaborative Filtering – A recommendation method that identifies similarities among users to predict which marathon events a runner may prefer based on the interests of similar individuals.

Data Preprocessing – The process of cleaning, validating, and organizing user and event data to ensure accuracy before it is processed by the system.

Event Data – Information related to marathon events, such as schedules, locations, categories, fees, and available slots.

Hybrid Recommendation Algorithm – A combined recommendation method using both content-based and collaborative filtering approaches to provide more personalized and accurate marathon event suggestions.

Iterative SDLC Model – A Software Development Life Cycle methodology in which system development occurs in repeated cycles of planning, designing, building, and testing to gradually improve system features.

Marathon Race Management – A framework that covers key areas of marathon event preparation and execution, including communication, service management, environment and entertainment, medical services, and post-race engagement.

Postrace Engagement – Activities and services provided after the marathon (e.g., results, medals, feedback gathering) to improve runner satisfaction and community involvement.

Registration Slot Allocation – The process of assigning available marathon slots to participants based on specific criteria such as registration time, categories, and priority rules.

Runner Profile – A collection of information about a runner, including age, gender, running experience, preferred distance, and event location preferences.

Software Development Life Cycle (SDLC) – A structured process used to plan, design, build, test, and deploy the system to ensure it meets user and technical requirements.

User Authentication – A system process that verifies the identity of a user (runner, organizer, or admin) before granting access to the platform.

Weighted Matching Algorithm – An optimization method that assigns marathon slots fairly based on predefined weighted criteria such as runner category, registration time, and event capacity

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a comprehensive review of related literature and studies that inform the design and development of RunConnect, a digital platform intended to improve marathon registration, participant engagement, and event management. It examines foreign and local literature on marathon registration systems, user experience issues, event logistics, and algorithm-based solutions. Furthermore, it discusses both foreign and local academic studies related to recommendation systems, slot allocation, and sports event management, establishing the academic foundation for the proposed system.

Foreign Literature

Foreign literature highlights the continuous challenges faced by international marathon events in managing registration systems and participant workflows. Numerous studies and industry reports indicate that large-scale marathons commonly experience system overloads, long queues, and participant dissatisfaction due to outdated or poorly optimized registration platforms. These issues manifest as system failures, unclear communication, incomplete entries, and poor user experience, ultimately affecting participant satisfaction and conversion rates [18]. Even well-established events continue to struggle with sudden surges in registration volume, particularly during peak enrollment periods.

Research on marathon risk and logistics further emphasizes the operational complexity of running events. Large-scale marathons face risks such as crowd surges, inefficient coordination, and delayed communication, which underscore the need for digital platforms that support automated processes and data-driven decision-making [19]. These findings align with the ongoing global shift toward digital transformation in sports event management.

Additionally, international studies on participant satisfaction reveal that marathon runners increasingly expect convenient, user-friendly, and efficient online systems. Huang (2024) reports that marathon participants in Taiwan express dissatisfaction with registration processes due to payment issues, refund complications, and confusing sign-up procedures [20]. These challenges indicate that registration inefficiencies are not limited to developing countries but are widespread across global events.

Collectively, foreign literature demonstrates the need for smarter, more reliable, and more personalized marathon registration platforms. This supports the relevance of RunConnect as an innovative system integrating recommendation algorithms and fair slot allocation mechanisms to address persistent issues in international marathon events.

Local Literature

Local Philippine literature and reports likewise highlight the challenges faced by marathon organizers in adopting efficient digital systems. As running events continue to grow nationwide, local organizers still rely heavily on traditional or semi-digital registration methods. These methods often involve manual verification, physical form checking, or basic online forms, which lead to delays, inconsistencies, and incomplete submissions. For example, the MassKara Marathon registration process continues to involve manual components, creating inefficiencies in participant processing and verification [21].

Similarly, online running directories such as RunPH serve primarily as listing platforms without advanced features like personalized recommendations, predictive filtering, or automated slot allocation. Runners must manually browse through extensive event lists, as the platform does not employ intelligent algorithms to streamline user experience [22]. This limitation reflects the broader issue of insufficient digital innovation in local sports event management.

Local academic studies also reveal that many Philippine event organizers struggle with adopting fully automated platforms due to limited technical training, lack of

digital infrastructure, and reliance on manual data management. Dela Cruz and Reyes (2022) report that large-scale event systems in the Philippines continue to encounter challenges related to automation, data handling, and participant monitoring [23]. These findings emphasize the need for more adaptive and technology-driven solutions in local marathon and sports event management systems.

Foreign Studies

Foreign studies provide strong empirical support for the application of hybrid recommendation systems and matching algorithms in marathon event management. Researchers highlight the growing integration of machine learning and recommender systems in sports technologies, where algorithms are increasingly used to suggest training plans, events, and motivational content based on runners' performance and preferences [24]. Another study proposes a predictive system that recommends challenging yet achievable race goals for marathoners, demonstrating the feasibility and value of algorithm-driven event personalization.

Moreover, the integration of demographic-based performance benchmarks is essential for accurate runner classification and safety verification. Quantitative analyses of global running data indicate that defining a 'good' pace is strictly dependent on demographic variables such as age and gender. According to RunningPace (2025), average finishing times vary significantly across different cohorts for instance, the statistical baseline for an 'Intermediate' runner in the 20-29 age bracket is markedly faster than that of the 50-59 bracket [25]. This literature suggests that for a system to effectively verify a runner's capability rather than relying on subjective self-assessment it must utilize a dynamic comparison mechanism that cross-references a runner's historical time against these global age-graded standards. This validates the necessity of using data-driven logic to filter applicants for elite slots or strict cut-off categories.

Optimization and algorithmic literature also show that hybrid or advanced classification models can increase fairness and efficiency in event systems. Studies propose hybrid classification approaches that balance user interests with event

constraints, such as limited slots or varying difficulty levels [26]. Furthermore, evidence suggests that combining recommendation algorithms with matching or optimization techniques creates more balanced and equitable systems for users [27].

These foreign studies validate the theoretical foundation of RunConnect, particularly its combined use of hybrid recommendation and weighted matching algorithms to deliver personalized suggestions and fair slot allocation.

Local Studies

In the Philippine setting, only a limited number of academic studies focus directly on marathon registration systems however, practical observations reveal persistent inefficiencies. Prominent race organizers such as RUNRIO continue to use basic online registration forms without algorithm-based tools to manage demand or prevent overbooking. For example, the Rock ’n’ Roll Manila Marathon registration page indicates limited slots and non-refundable entries, which can lead to user frustration during peak registration periods.

Likewise, platforms such as RunPH list hundreds of running events but do not provide personalized recommendations or automated matching features, requiring runners to manually search through event lists. This highlights a significant gap: the absence of intelligent, data-driven registration systems in the country’s marathon industry.

These limitations in existing local systems support the development of RunConnect, which aims to introduce a more advanced platform integrating personalized recommendations, optimized slot allocation, and automated registration processes tailored to the needs of Filipino runners and organizers.

Literature Map

Table 1. Foreign Literature Related to Marathon Registration and Event Management

Author / Year	Focus of the Study	Key Findings	Relevance to RunConnect
Axelsen & Robinson (2009)	Distance running research agenda	Identified growth and complexity of marathon participation worldwide	Supports the need for structured and scalable marathon systems
Hall et al. (2019)	Stakeholder risks in marathon events	Highlighted coordination, safety, and management challenges	Justifies integrated digital platforms for coordination
World Athletics (2021)	Marathon event guidelines	Emphasized communication, safety, and service management	Provides theoretical basis for system requirements
Huang & Chiu (2022)	Participant satisfaction in marathons	Reported dissatisfaction with slow and inefficient registration systems	Reinforces need for optimized registration workflows
Phua et al. (2024)	Digital technology in marathon services	Highlighted that integrating real-time monitoring and digital checkpoints significantly reduces safety risks and	Provides the empirical basis for the "Geospatial Tracking Engine," validating that real-time

		improves logistical coordination during races.	visualization is essential for modern race safety.
RunningPace.io (2025)	Running Pace Benchmarks and Demographic Performance Standards	Performance standards are strictly relative to Age and Gender. Data confirms that average finishing times vary significantly by age group, making universal standards inaccurate.	It provides the scientific basis for the "Pace Calculation Engine," justifying the use of demographics to automatically verify runner capability instead of relying on self-assessment.

Table 2. Local Literature Related to Marathon Registration Systems

Author / Organization / Year	System or Context	Identified Issues	Implications for the Study
MassKara Sports Committee (2024)	MassKara Marathon registration	Manual verification and partial digital processing	Highlights inefficiency of semi-digital systems
RunPH Platform	Online event	No personalization or	Shows gap in

	listing platform	automated slot allocation	intelligent recommendation features
Dela Cruz & Reyes (2022)	Philippine sports event systems	Limited automation and data handling challenges	Supports need for fully automated platforms

Table 3. Foreign Studies on Recommendation and Matching Algorithms

Author / Year	Algorithm / Method	Application Area	Contribution to RunConnect
Smyth & Cunningham (2017)	Recommender system	Runner Classification & Safety	Basis for hybrid recommendation design
Afoudi et al. (2021)	Hybrid recommendation system	Personalized recommendations	Validates combining content-based and collaborative filtering
Smyth et al. (2022)	Machine learning recommender systems	Runner support technologies	Supports accuracy and personalization objectives

Nithya et al. (2022)	Hybrid classification model	Event recommendation systems	Supports effectiveness of hybrid models

Table 4. Algorithmic and Optimization Studies Related to Slot Allocation

Author / Year	Theory / Algorithm	Key Concept	Relevance to the Study
Edmonds (1965)	Matching theory	Optimal and fair matching	Foundation of weighted matching algorithm
Nithya et al. (2024)	Optimization models	Fair and balanced allocation	Supports transparent slot distribution

Table 5. Summary Literature Map (Research Gap Identification)

Category	Existing Literature Findings	Identified Gap	RunConnect Contribution
Registration Systems	Mostly manual or basic online forms	Lack of automation and intelligence	Automated, algorithm-driven registration

Personalization	Limited or no recommendations	One-size-fits-all event listing	Hybrid recommendation algorithm
Slot Allocation	First-come-first-served approach	Unfair during peak registration	Weighted matching algorithm
User Engagement	Minimal interaction features	Low engagement and satisfaction	Interactive dashboards and recommendations

CHAPTER III

METHODOLOGY

This chapter outlines the procedures, techniques, and methodological processes that will guide the development of *RunConnect*, an intelligent marathon registration and runner event matching system. The methodology will describe how the research will be conducted, what system components will be developed, and how the platform will be evaluated. This chapter will also justify the selected research methods to ensure that the study will be replicable and scientifically grounded.

Research Design

This study will utilize a Developmental Research Design, which will focus on the systematic design, development, and evaluation of an instructional or technological product. This design is appropriate as the primary goal is to construct a new artifact the RunConnect system integrated with novel algorithms. The development process will follow the Iterative SDLC Model, allowing for continuous refinement of the Predictive Analytics and Geospatial Tracking modules. Furthermore, a Descriptive Evaluative Method will be employed post-development to assess the system's "Intelligent" capabilities, specifically measuring the accuracy of the category recommendations against actual runner performance.

Materials

The development, testing, and deployment of the RunConnect system require appropriate hardware, software, data resources, and human support to ensure reliable system performance and accurate evaluation. These materials provide the technical foundation necessary for implementing the hybrid recommendation and weighted matching algorithms, conducting usability and performance testing, and validating the system in accordance with the study's objectives.

Table 6. Hardware Requirements for the Development and Evaluation of RunConnect

Hardware Component	Specification	Purpose
Development Computer	Intel Core i5 processor or higher, 8 GB RAM, sufficient storage	Used for system coding, integration, testing, and deployment
Internet Connectivity	Stable broadband connection	Supports development, cloud hosting access, documentation, and collaboration
Testing Devices	Desktop computers, laptops, tablets, smartphones	Used for cross-device compatibility and responsive design testing
Web Server / Cloud Hosting	Shared or cloud-based hosting environment	Hosts the system during testing and evaluation

Table 7. Software and Data Requirements for the Development and Evaluation of RunConnect

Software / Resource	Description	Purpose
Operating System	Windows, macOS, or Linux	Platform for development and deployment
Local Server Environment	XAMPP, WAMP, or Laravel Valet	Enables local backend development and testing
Backend Framework	Laravel (PHP)	Implements system logic, authentication, and data processing
Programming Languages	PHP, JavaScript	Backend development, algorithm implementation, and frontend interactivity

Frontend Technologies	HTML, CSS, JavaScript	User interface and client-side functionality
Recommendation Libraries	PHP	Development and evaluation of recommendation algorithms
Database Management System	MySQL	Stores user profiles, events, and registration records
Development Tools	Code editors and database management tools	Debugging, optimization, and system maintenance
Datasets	Fictional user profiles, marathon events, and simulated registrations	Used for testing system functionality and recommendation accuracy
Documentation and References	Technical manuals and academic literature	Guides system development and algorithm design

The hardware and software resources listed in Tables 6 and 7 provide the necessary infrastructure for the effective development, testing, and deployment of the RunConnect system. A development computer with adequate processing power and memory ensures efficient handling of backend operations, algorithm execution, and database transactions. The availability of multiple testing devices enables thorough evaluation of system responsiveness and usability across different platforms and screen sizes, which is essential for a web-based application intended for diverse users.

The selected software stack supports scalable and structured system development. Laravel and PHP facilitate secure backend implementation, while PHP and its associated libraries enable the execution and evaluation of the hybrid recommendation and weighted matching algorithms. Standard web technologies ensure

a responsive and user-friendly interface, and MySQL provides reliable data storage for user, event, and registration information. The use of fictional datasets allows controlled testing without compromising data privacy, while technical documentation and academic references ensure that development follows established best practices. Collectively, these materials support the accurate evaluation and validation of RunConnect in alignment with the study's technical and research objectives.

Model Implementation

The development of RunConnect will follow a structured process that integrates data collection, algorithm design, system implementation, and testing. This section outlines the specific procedures and techniques that will be employed to build the core components of the platform, particularly the hybrid recommendation algorithm and the weighted matching algorithm. The model development process consists of several interconnected phases, beginning with data collection and proceeding through preprocessing, algorithm implementation, and system integration.

Data Gathering Procedure

Data collection serves as the critical foundation for the RunConnect system, feeding both the Machine Learning recommendation engine and the Real-Time Safety Module. The process begins with User Profiling, where participants provide basic demographics (age, gender) and self-reported running preferences. These inputs serve as the primary variables for the system's intelligent category recommendations, allowing for immediate registration without manual document validation.

Simultaneously, event organizers input Geospatial Data, defining the race route coordinates and safety geofences rather than just static checkpoints. Finally, data gathering extends to the actual event through Real-Time GPS Tracking. Instead of relying solely on manual checkpoint scanning, the system captures Location Logs (Latitude, Longitude, and Timestamps) from the runner's device. This stream is immediately processed to detect off-route anomalies and is stored in the Event DB to continuously refine the runner's future category recommendations.

Data Preprocessing and Data Cleaning

Once raw data has been collected, it must undergo systematic cleaning and processing to ensure accuracy and consistency.This phase will prepare all data for use in the recommendation and matching algorithms.

The system will implement validation rules to detect and correct common data entry errors. For user data, this includes checking for valid email formats, realistic age ranges, correct phone number formats, and logical consistency in running experience. For event data, the system will validate that event dates are in the future, registration deadlines occur before event dates, slot numbers are positive integers, and locations match standardized city and region names. When errors are detected during data entry, users will receive immediate feedback to correct the information. Existing records will be scanned periodically to identify and flag any anomalies for review.

Duplicate entries will be identified and removed to prevent unfair slot allocation and data inconsistencies. The system will check for duplicate user accounts based on email addresses and phone numbers, preventing users from creating multiple accounts with the same contact information. For event registrations, the system will ensure that each user can only register once per event category. Event listings will also be checked for duplication based on event name, date, and location to avoid confusion. Database constraints will enforce these rules to prevent duplicates from being created.

Missing data will be handled based on field importance. Mandatory fields such as email, event name, event date, and available slots must be completed before submission. For optional fields like preferred marathon distance or terrain type, the system will either assign default values or mark them as unspecified. New users with limited interaction history will rely more on content-based filtering until sufficient behavioral data is collected for collaborative filtering.

All data will be formatted according to consistent standards. Dates will use YYYY-MM-DD format, locations will follow a standardized hierarchy (venue, city, region, country), and categorical variables like experience level (beginner, intermediate, advanced, elite)

will use controlled vocabularies. Numerical values will maintain consistent precision, with distances in kilometers and fees in decimal format. This standardization ensures that the algorithms can process data efficiently and accurately.

To support algorithmic processing, certain data will be transformed into numerical representations. User preferences and event characteristics will be encoded into feature vectors for the recommendation algorithm. Categorical variables will be converted using encoding techniques, and priority weights for the matching algorithm will be normalized to ensure all factors contribute proportionally to slot allocation decisions.

Model Selection

The model selection process for RunConnect focuses on addressing personalization and fair slot allocation in marathon registration. A hybrid recommendation algorithm combining content-based and collaborative filtering will be implemented to generate personalized event suggestions by matching user preferences with event attributes while also leveraging behavioral patterns from similar users, thereby overcoming limitations such as narrow recommendations and cold-start issues. In parallel, a weighted matching algorithm will be used for slot allocation to ensure a transparent, consistent, and rule-based process. This algorithm assigns predefined weights to criteria such as registration time, runner category, membership status, and event capacity, computes priority scores for each registration request, and allocates slots accordingly. Together, these models support accurate recommendations and fair, explainable registration decisions aligned with the practical requirements of marathon event management.

Model Training

Model training in RunConnect focuses on calibrating the hybrid recommendation algorithm using prepared training data, while the weighted matching algorithm does not require training due to its rule-based nature. The content-based component is trained by learning the relationships between user profiles and event characteristics such as distance, location, difficulty, and terrain, with feature weights adjusted according to their

influence on user interest. The collaborative filtering component is trained using user interaction data to identify similarities among users and predict preferences based on shared registration patterns, employing similarity-based or matrix factorization techniques. Both components are combined using a weighted approach, initially balanced and later adjusted during validation to address cold-start users and improve accuracy. Key parameters, including feature weights, similarity thresholds, and the balance between filtering methods, are optimized through iterative training cycles using metrics such as precision, recall, and recommendation diversity until stable and acceptable performance is achieved.

Statistical Treatment

The collected data from system testing and user feedback will be analyzed using appropriate statistical methods to evaluate the performance, accuracy, and usability of RunConnect. For quantitative data, descriptive statistics including mean, median, and standard deviation will be computed to summarize user satisfaction ratings, system performance metrics, and algorithm accuracy scores. Frequency distribution and percentage analysis will be used to categorize respondent demographics, experience levels, and preference patterns. The effectiveness of the hybrid recommendation algorithm will be measured using precision, recall, and F1-score metrics, which quantify how accurately the system suggests relevant events to users. The fairness of the weighted matching algorithm will be evaluated by analyzing the distribution of slot allocations across different priority categories to verify that assignments follow predefined criteria without bias or errors.

Table 8. Data Analysis Procedures

Evaluation Aspect	**Instrument / Data Source**	**Measurement Scale / Method**	**Analysis Technique**	**Tools Used**

User Satisfaction	Survey Questionnaire	5-point Likert Scale	Weighted Mean computation and interpretation	Microsoft Excel
Usability Evaluation	Survey Questionnaire	5-point Likert Scale	Descriptive statistics (Weighted Mean)	Microsoft Excel
Ease of Navigation	Survey Questionnaire	5-point Likert Scale	Weighted Mean computation	Microsoft Excel
Registration Efficiency	Survey Questionnaire	5-point Likert Scale	Weighted Mean computation	Microsoft Excel
Recommendation Relevance	Survey Questionnaire	5-point Likert Scale	Weighted Mean computation	Microsoft Excel
Algorithm Accuracy	System Logs / Test Data	Precision, Recall, F1-Score	Confusion Matrix Analysis	PHP

Table 9. Likert Scale Interpretation for User Satisfaction

Mean Range	Verbal Interpretation
4.50 – 5.00	Strongly Satisfied
3.50 – 4.49	Satisfied
2.50 – 3.49	Neutral
1.50 – 2.49	Dissatisfied
1.00 – 1.49	Strongly Dissatisfied

System Development Methodology

The researchers will utilize the Iterative System Development Life Cycle (SDLC) Model for the development of *RunConnect*. This methodology is selected because it allows the system to be developed in repeated cycles (iterations), enabling the gradual integration and refinement of complex features such as the Predictive Analytics Module and Geospatial Tracking Engine. Unlike the traditional Waterfall model, the Iterative Model allows the researchers to revisit previous phases to improve the system based on testing results. The development process is divided into the following phases:

1. Planning and Requirement Analysis Phase In this initial phase, the researchers will gather the necessary data and define the scope. This involves:

- Collecting the Training Dataset (n=100+) through surveys to serve as the baseline for the machine learning model.
- Defining the logical rules for the Weighted Matching Algorithm, such as specific age constraints and health safety checks.
- Identifying the technical requirements, including the browser-based Geolocation API capabilities for tracking.

2. Design Phase The researchers will translate the requirements into system blueprints. This includes:

- Designing the System Architecture to separate the Presentation, Application, and Data layers.
- Creating the Entity-Relationship Diagram (ERD) to structure the database for user profiles, location logs, and event details.
- Designing the specific algorithms (Decision Tree and Weighted Matching) using flowcharts and pseudocode.

3. Development and Implementation Phase (The Iterations) This phase involves the actual coding of the system, executed in iterative cycles:

- Iteration 1 (Core Features): Development of the web interface, user registration forms, and database connectivity.
- Iteration 2 (Intelligence): Implementation of the Pace Calculation Engine and the Predictive Analytics Module to process user data and generate recommendations.
- Iteration 3 (Tracking): Integration of the Geospatial Tracking Service to enable real-time map visualization and geofence alerts.

4. Testing and Validation Phase Each iteration undergoes rigorous testing. The researchers will conduct:

- Unit Testing: To verify if individual components (e.g., pace calculator) function correctly.
- Algorithm Testing: To measure the accuracy of the recommendations against the training dataset using statistical metrics.
- System Testing: To evaluate the responsiveness of the GPS tracking feature under different mobile network conditions.

5. Evaluation Phase Upon completion, the system will be evaluated by end-users (runners and organizers) using a standardized questionnaire based on ISO 25010 to assess satisfaction and usability. Feedback from this phase will determine if further iterations are needed.

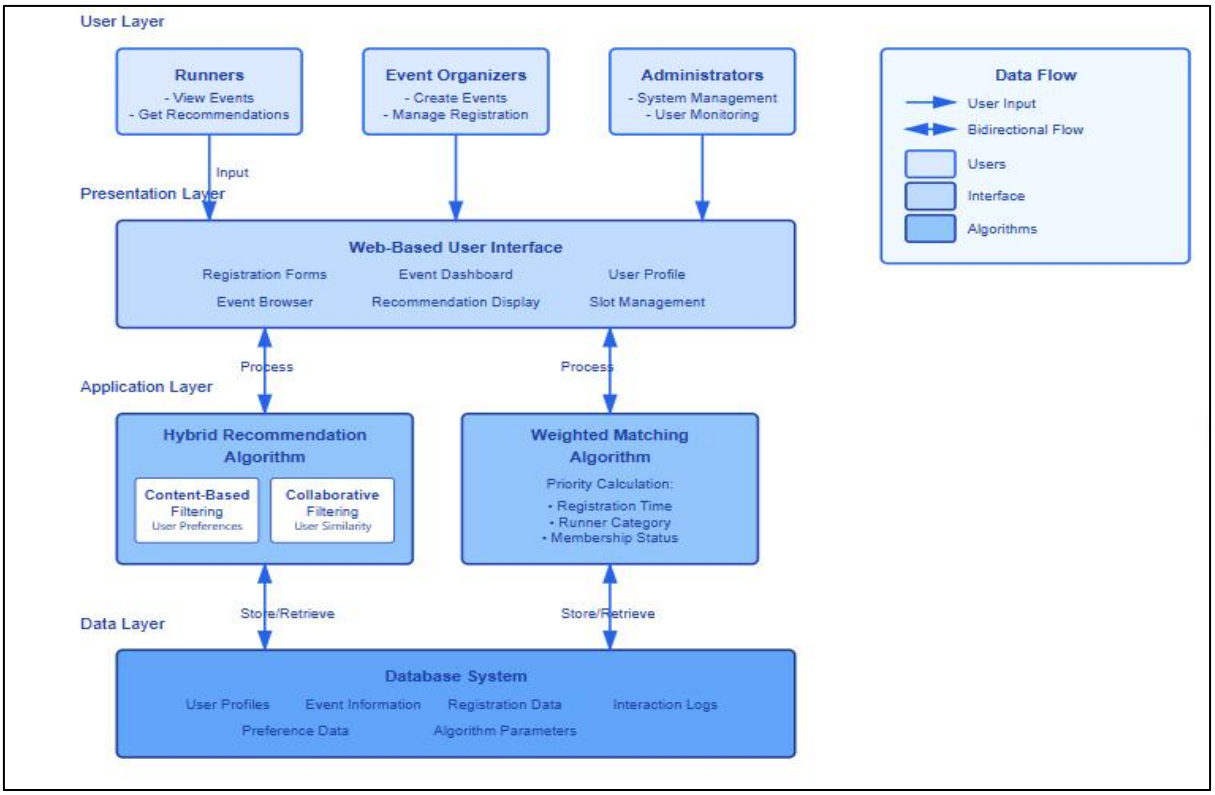


Figure 6. System Architecture of Block Diagram [7]

Figure 6 illustrates the system architecture of RunConnect, which is designed using a layered structure to ensure efficiency, scalability, and clear separation of functions. The architecture consists of four main layers: the user layer, presentation layer, application layer, and data layer. Runners, event organizers, and administrators interact with the system through a web-based user interface in the presentation layer, where user inputs and requests are collected. These requests are processed in the application layer, which contains the hybrid recommendation algorithm for personalized event suggestions and the weighted matching algorithm for fair slot allocation. All user information, event data, registration records, and algorithm parameters are securely stored and managed in the data layer. This layered architecture enables organized data flow, simplifies system maintenance, and supports reliable marathon registration and event management.

System Evaluation

The System Evaluation phase assesses the effectiveness, usability, accuracy, and performance of the developed system, including its embedded algorithms. Evaluation was conducted through system testing and user feedback involving runners, event organizers, and administrators.

User satisfaction and usability were measured using a 5-point Likert scale questionnaire, focusing on system features such as ease of navigation, registration efficiency, and recommendation relevance. Algorithm performance was evaluated by analyzing the accuracy and effectiveness of the hybrid recommendation algorithm and the fairness and efficiency of the weighted matching algorithm during slot allocation.

Quantitative data were analyzed using descriptive statistical methods, while qualitative feedback from interviews and open-ended survey responses was analyzed thematically to identify recurring issues and areas for improvement. The results of this evaluation served as the basis for validating the system's overall performance and reliability.

System Design

The system design phase presents the technical design and structure of RunConnect through standardized modeling diagrams that illustrate how all components work together.

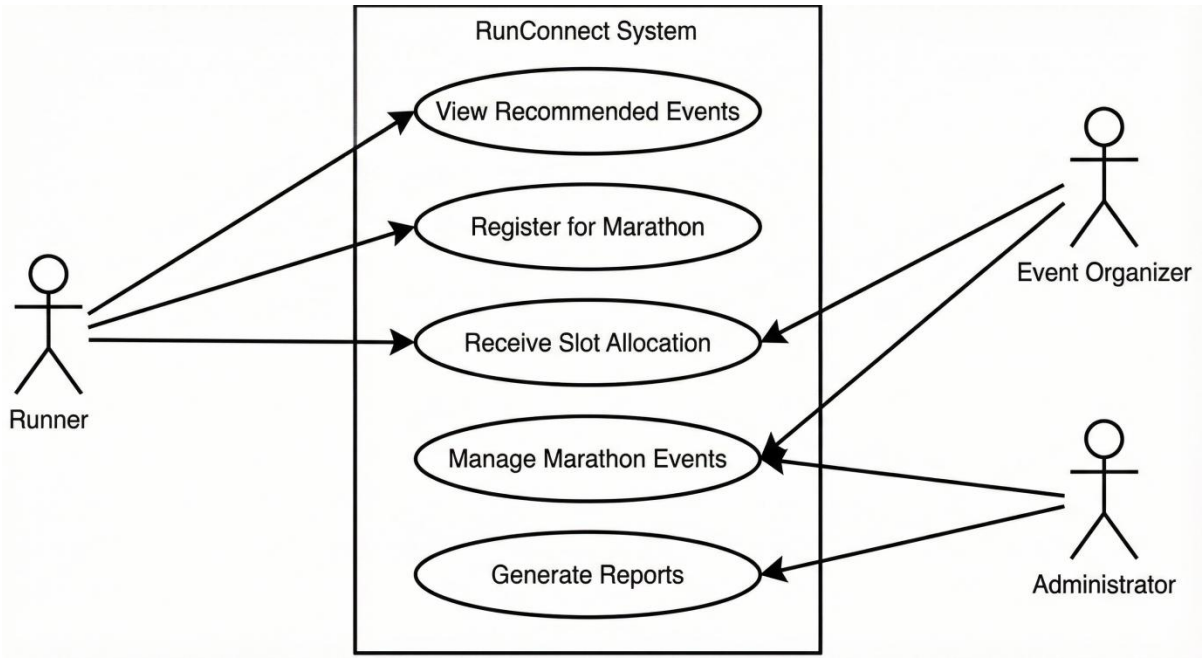


Figure 7. Use Case Diagram [7]

As shown in Figure 7, it illustrates the use case diagram of the RunConnect system, showing the main actors and their interactions with the platform. The system involves three key users: the Runner, Event Organizer, and Administrator. Runners use the system to view personalized marathon recommendations, register for events, and receive slot allocation results generated through the hybrid recommendation and weighted matching algorithms. Event organizers interact with the platform to create, update, and manage marathon events, monitor registrations, and oversee slot availability. The administrator is responsible for system-wide management, including event oversight, report generation, and maintaining data integrity and access control. Overall, the use case diagram provides a clear overview of how RunConnect supports personalized event discovery, fair slot allocation, and efficient marathon management through well-defined user roles and system functions.

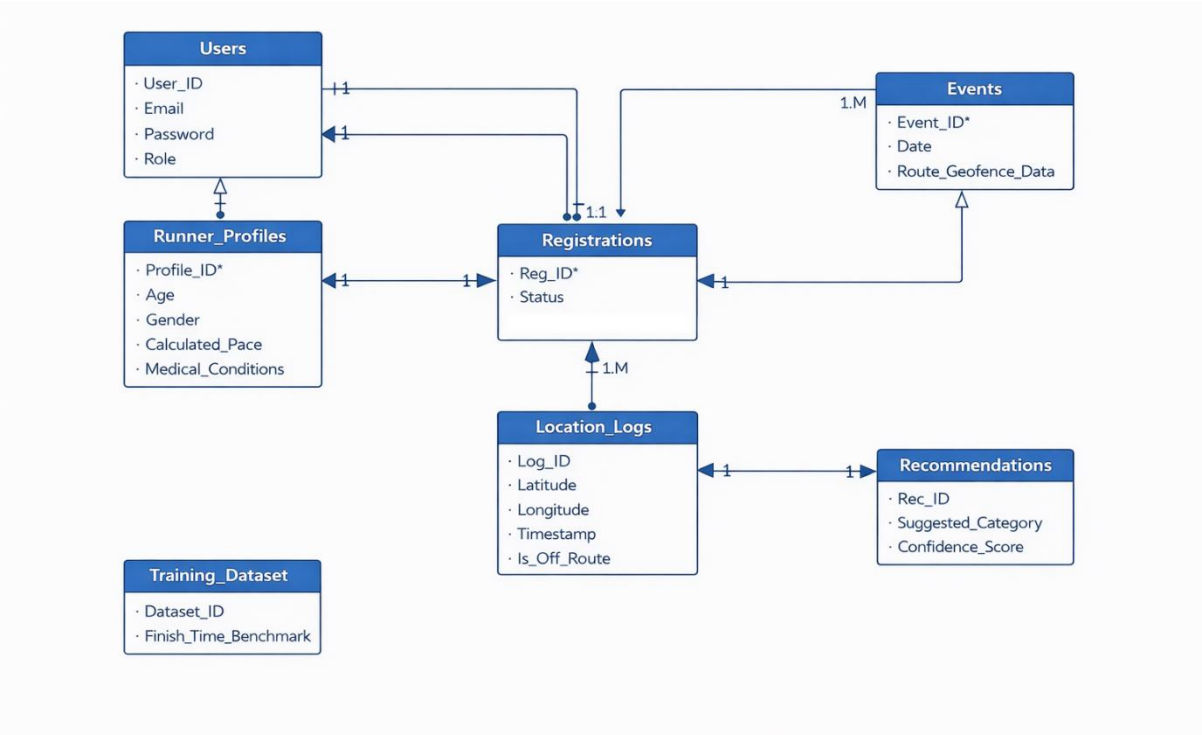


Figure 8. Entity Relationship Diagram (ERD) [8]

Figure 8 depicts the Entity-Relationship Diagram (ERD) of RunConnect, organizing the database into seven core entities. This relational schema bridges standard user data with specialized components for Machine Learning (Training_Dataset) and geospatial tracking (Location_Logs). The design guarantees data consistency, enabling the system to efficiently execute both intelligent category recommendations and real-time runner monitoring.

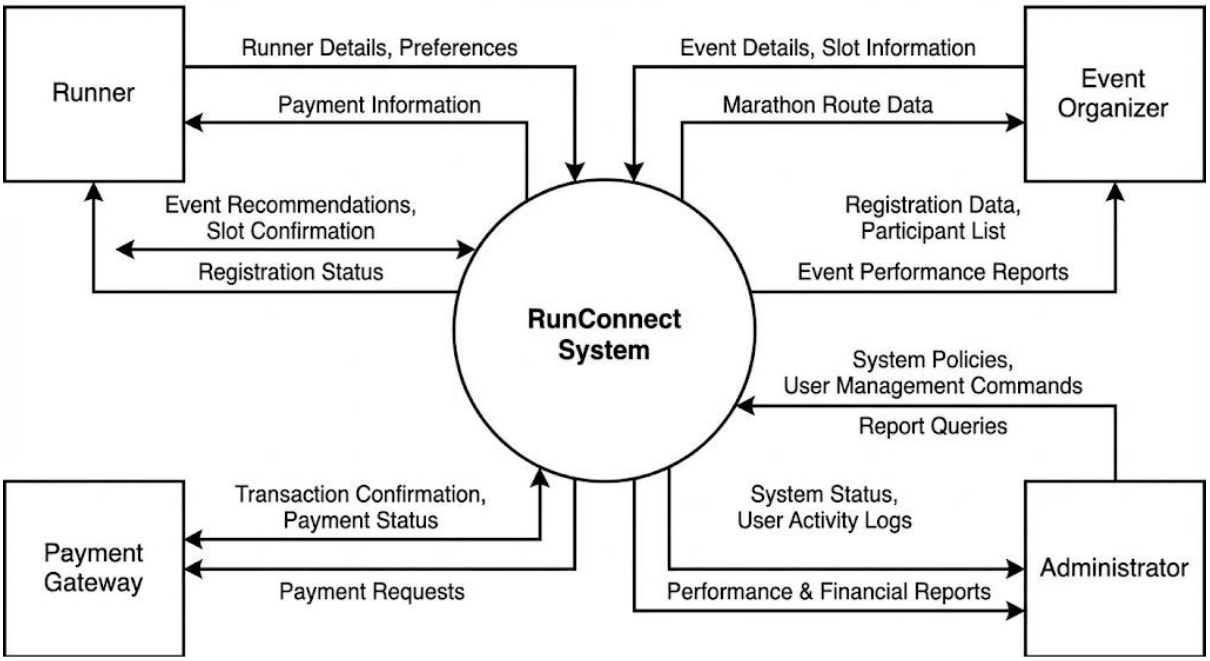


Figure 9. Data Flow Diagram Level 0 [9]

Context Diagram Figure 9 illustrates the Context Diagram (Level 0) of the RunConnect system, defining the project's scope and boundaries as a single entity interacting with four external entities. Runners interact with the system to input preferences and receive event recommendations, while Event Organizers provide route data and receive performance reports. Administrators oversee system policies and financial reports, and an external Payment Gateway is integrated to securely process transaction requests and return payment status, addressing the need for efficient transaction handling.

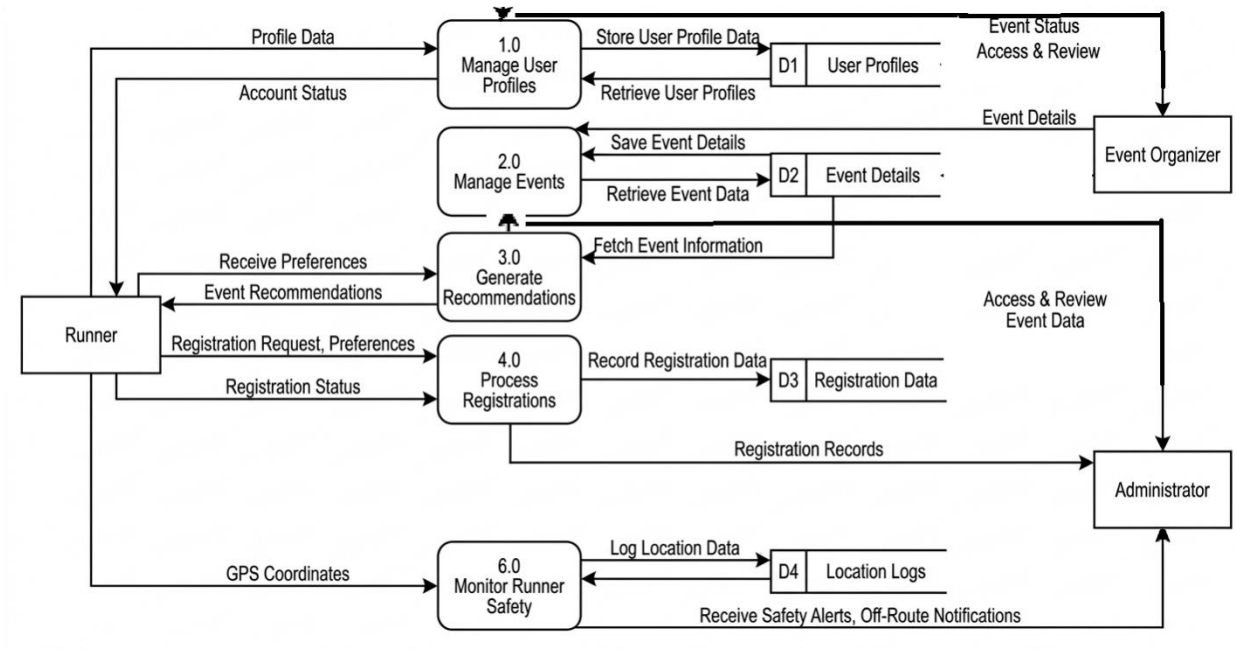


Figure 10. Data Flow Diagram Level 1 – Main Processes *[10]*

Main Processes Figure 10 presents the Data Flow Diagram Level 1, expanding the system into six core functional processes that fulfill the study's specific objectives. The system flow begins with Managing User Profiles (1.0) and Events (2.0) to store demographic and slot information, which feeds into the Generate Recommendations (3.0) process using the Hybrid Recommendation Algorithm. The workflow continues through Process Registrations (4.0), where the Weighted Matching Algorithm checks slot availability before saving data, while the Generate Reports (5.0) process aggregates data for administration. Finally, the Monitor Runner Safety (6.0) process captures real-time GPS coordinates to log location data and trigger immediate safety alerts to organizers if a runner deviates from the route.

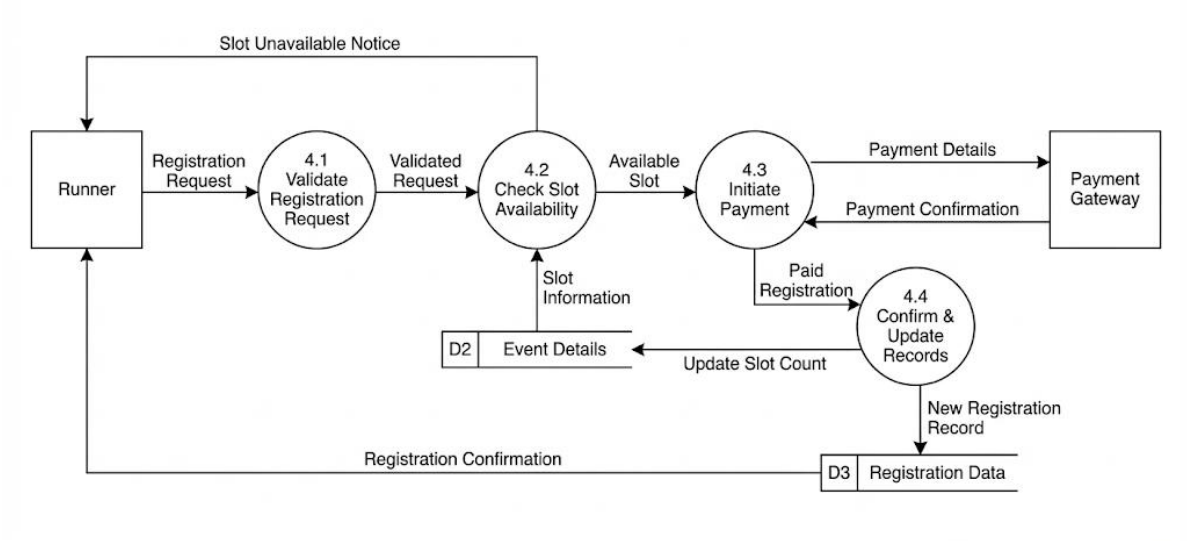


Figure 11. Data Flow Diagram Level 2 – Registration Sub-process *[11]*

Figure 11 details the sub-processes of the Process Registrations (4.0) module, illustrating the logic behind the system's fair slot allocation. The workflow initiates with validating the registration request (4.1) against age and category requirements, followed by checking slot availability (4.2) using the Weighted Matching Algorithm to verify priority-based capacity. Once a slot is confirmed available, the system initiates payment (4.3) through the external gateway and concludes by confirming and updating records (4.4), ensuring that slots are allocated fairly and overbooking is prevented before any financial transaction occurs.

Registration Workflow Model

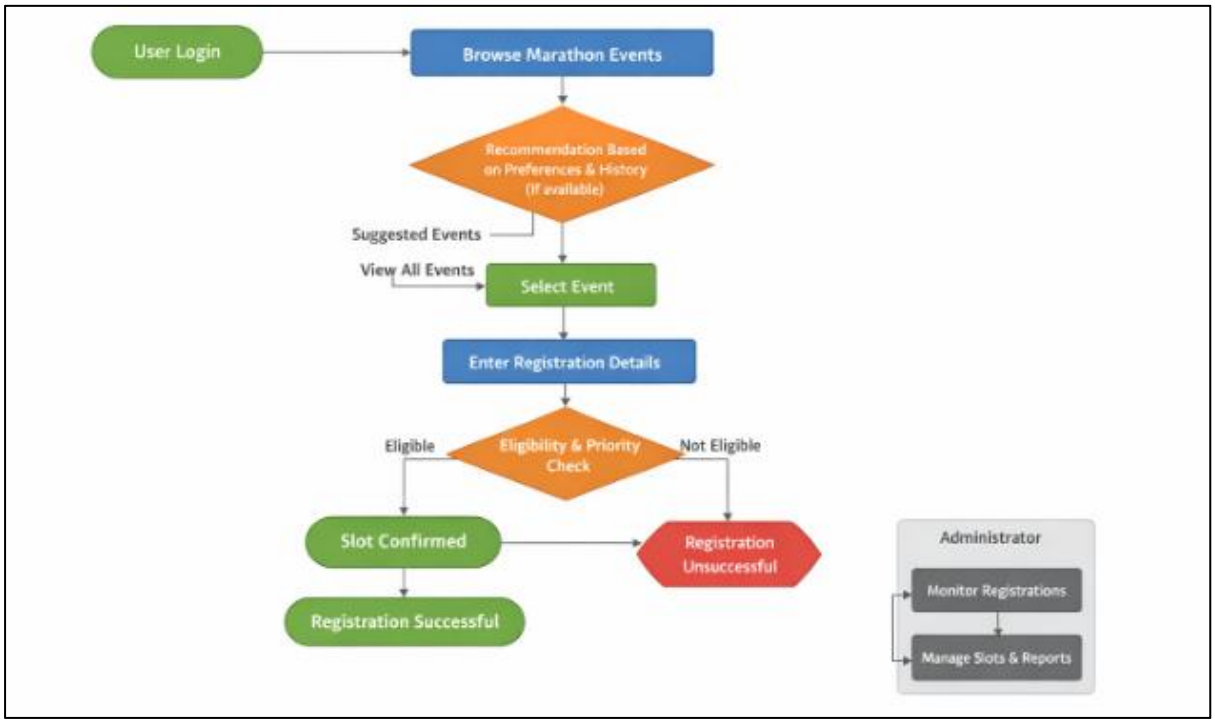


Figure 12. Registration Workflow Diagram of RunConnect [12]

The Registration Workflow Model defines the step-by-step process followed by users during marathon event registration. It illustrates how runners interact with the system from account access to final slot confirmation, ensuring an efficient and fair registration process.

The workflow begins when a runner logs into the system and browses available marathon events. Based on user preferences and historical participation data when available, the hybrid recommendation algorithm suggests relevant events. Once an event is selected, the runner proceeds to registration, where the weighted matching algorithm evaluates event-specific eligibility and priority criteria to allocate slots fairly. Successful registrations are confirmed and stored in the system database, while unsuccessful attempts are logged for monitoring and evaluation. Event organizers and administrators can oversee registrations, manage slots, and generate reports through the administrative dashboard.

This workflow ensures transparency, efficiency, and equitable slot distribution throughout the registration process.

Analytics Model Development

The analytics model development phase of RunConnect focuses on designing and implementing the hybrid recommendation algorithm and the weighted matching algorithm. These models were developed to provide personalized marathon event suggestions and ensure fair and efficient slot allocation during registration. The development process incorporated historical registration data, runner preferences, and eligibility criteria to optimize the accuracy and relevance of system recommendations while maintaining equitable slot distribution.

Validation and Testing of Models

The developed analytics models, including the hybrid recommendation algorithm and the weighted matching algorithm, were systematically validated and tested to ensure accuracy, reliability, and effectiveness. Data model validation verified the completeness, consistency, and correctness of all input data, including runner profiles, event details, and historical registration records. Process model validation ensured that the algorithms operated according to the intended workflows, properly executing event recommendations, eligibility checks, and slot allocations. Analytics validation evaluated the performance of the hybrid recommendation algorithm in providing accurate and relevant event suggestions and assessed the fairness and efficiency of the weighted matching algorithm in distributing registration slots. Validation involved simulation runs, comparison with expected outcomes, and feedback from runners, event organizers, and administrators, guaranteeing that the models functioned as intended and supported the overall usability, performance, and credibility of the RunConnect system.

Confusion Matrix

Precision: Of all runners the AI labeled as capable of '21km', how many were actually capable of running 21km?

$$Precision = \frac{TP}{TP + FP}$$

Recall (Sensitivity): Of all the actual '21km' runners, how many did the AI successfully identify/recommend?

$$Recall = \frac{TP}{TP + FN}$$

F1-Score: The balance between Precision and Recall. This is the most important metric for your study.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the results of the RunConnect study. The results follow the sequence of the four specific objectives: data collection, machine learning model design and training, geospatial tracking module design, and system evaluation. Each section presents the evidence produced for each objective, including collected datasets, algorithm outputs, simulation results, and user evaluation scores. The data comes from three real sources: the Dagan sa Hinatuan 2025 marathon (598 registrants), the Run for a Cause 2026 held at Sitio Maputi (153 documented runners), and an online survey (131 usable responses). The combined dataset totals 818 records.

Collection of Runner Demographics and Self-Reported Running History

The first specific objective required collecting runner demographics and self-reported running history. The researchers gathered data from three sources: the Dagan sa Hinatuan 2025 marathon online registration masterlist (598 runners), in-person interviews at Run for a Cause 2026 at Sitio Maputi (153 runners), and an online survey (131 usable responses). The combined dataset totals 818 records. This section presents the data sources, collection methods, demographic profile of participants, and the dataset split used for model training, validation, and testing.

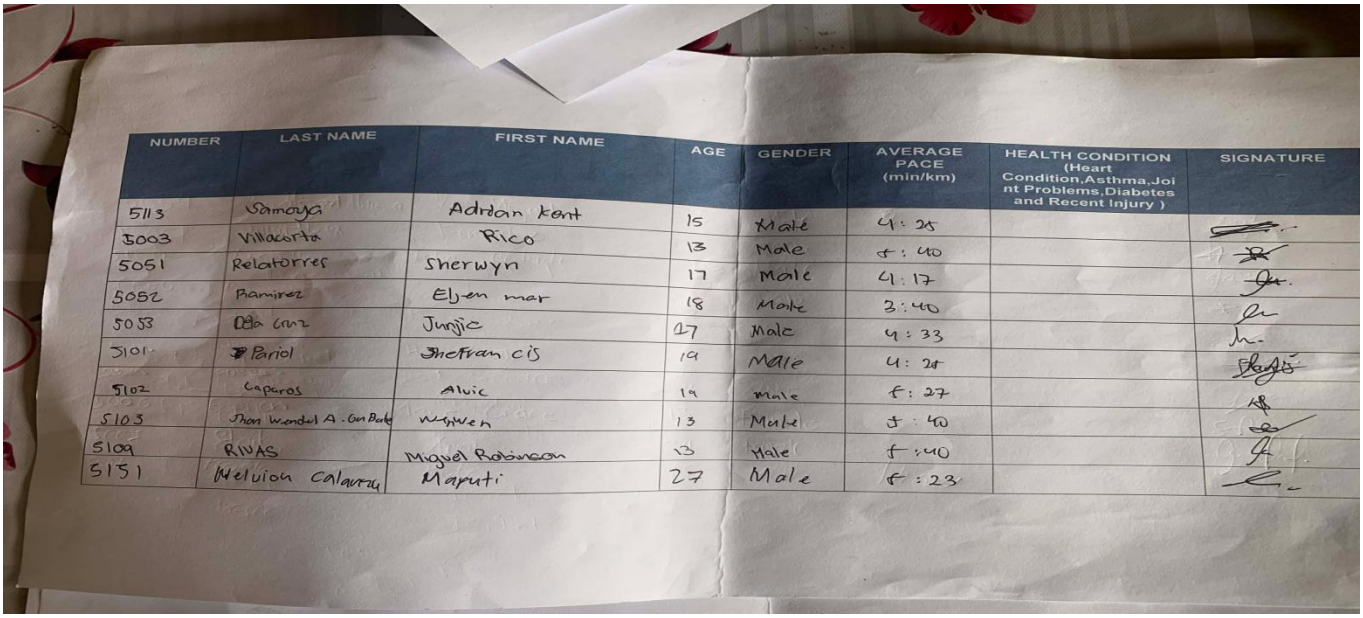
Data Collection Sources and Methods

Table 10. Data Collection Sources — Dagan sa Hinatuan 2025 and Run for a Cause 2026

Source	Method	Data Captured	Records Collected
Dagan sa Hinatuan 2025 — Official Online Registration (Primary Source)	Google Forms (online survey distributed by event organizer)	Age group, Gender, Running club/group, Category registered (5KM / 10KM / 21KM),	598 total registrants

Source	Method	Data Captured	Records Collected
		Shirt size, Emergency contact, Waiver acceptance	
Run for a Cause 2026 — In-Person Interview	Structured interview form (pace form) + race sign-up sheet conducted during race kit claim night	Runner number, Age, Gender, Average pace (min/km), Health condition (heart, asthma, diabetes, recent injury), Signature	153 total runners documented: 95 with self-reported pace, 58 confirmed 5KM finishers (no pace)
Online Survey — RunConnect Runner Profile Survey	Google Forms distributed via social media and running group chats	Age, Gender, Years of running experience, Usual race category, Number of events joined, Running club membership, Average pace range, Exact pace (optional), Best 5KM finish time (optional), Health conditions, Research consent	154 responses collected; 131 used (8 no consent, 15 no usable pace data)

Figure 13. Runners Participating in the In-Person Data Collection Session



NUMBER	LAST NAME	FIRST NAME	AGE	GENDER	AVERAGE PACE (min/km)	HEALTH CONDITION (Heart Condition,Asthma,Joint Problems,Diabetes and Recent Injury)	SIGNATURE
5113	Samoya	Adrian kent	15	Male	4:25		
5003	Villacorta	Rico	13	Male	5:40		
5051	Relatorres	Sherwyn	17	Male	4:17		
5052	Ramirez	Elyen mar	18	Male	3:40		
5053	Dela Cruz	Junjie	17	Male	4:33		
5101	Pariol	Shefran cis	19	Male	4:24		
5102	Caparos	Aluic	19	Male	5:27		
5103	Juan Wendel A. Gonbale	Nathen	13	Male	5:40		
5109	RIVAS	Miguel Robinson	13	Male	5:40		
5151	Meluion Calauru	Maruti	27	Male	5:23		

Figure 14. Sample Interview Data Collection Form Used at Run for a Cause 2026 (Names Withheld)

Dagan sa Hinatuan 2025 — Online Registration Dataset

The official online registration masterlist of Dagan sa Hinatuan 2025 served as the primary dataset for demographic profiling and machine learning model training. All participant names and contact information are withheld throughout this study to protect privacy. Runners are referenced by anonymized category groups in all analysis tables.

Table 11. Dagan sa Hinatuan 2025 — Registration Count by Category and Gender

Category	Total Registrants	Male	Female	Registration Period
5KM (Fun Run)	221	95	126	July – August 2025
10KM	199	90	109	July – August 2025
21KM (Half Marathon)	178	121	56 (1 unspecified)	July – August 2025
TOTAL	598	306	291 (+1)	July – August 2025

Table 12. Dagan sa Hinatuan 2025 — Age Group Distribution by Category

Age Group	5KM (n=221)	5KM %	10KM (n=199)	10KM %	21KM (n=178)	21KM %
19 and below	83	37.6%	—	—	—	—
20–29	—	—	75	37.7%	88	49.4%
30–39	37	16.7%	70	35.2%	55	30.9%
40 and above	50	22.6%	40	20.1%	31	17.4%

Age Group	5KM (n=221)	5KM %	10KM (n=199)	10KM %	21KM (n=178)	21KM %
Not specified	51	23.1%	14	7.0%	4	2.2%

Table 13. Dagan sa Hinatuan 2025 — Registration Timeline

Category	July 2025	August 2025	% in August	Observation
5KM	20	201	91.0%	Strong late-period surge
10KM	35	164	82.4%	Majority registered in August
21KM	38	137	78.3%	Even competitive category surged late
TOTAL	93	502	83.9%	83.9% of all registrations arrived in a single month

The registration timeline reveals a critical operational pattern: 83.9% of all 598 registrations (502 out of 598) arrived in August 2025 a single-month surge. This real-world observation directly motivates the Weighted Matching Algorithm in RunConnect. Without an automated priority-based slot allocation system, this surge would overwhelm manual registration management and create unfair outcomes for early registrants who committed to the event weeks earlier.

In-Person Interview Dataset Run for a Cause 2026

The in-person interview component was conducted by the researchers at the Run for a Cause 2026 event. Runners were approached during the race kit claim night and asked to fill out a structured interview form capturing their average pace per kilometer and health condition. A total of 153 runners were documented across two forms: (1) 95 runners completed the interview form and provided a self-reported average pace in min/km, and (2) 58 confirmed 5KM finishers signed the registration

sign-up sheet only and could not report their pace. All 95 interview runners are included in the ML training dataset with their real self-reported pace values. The 58 finishers without pace are included with pace imputed from age and gender using RunningPace.io (2025) standards, with their category fixed as 5KM since they are confirmed participants and finishers of the 5KM event at Run for a Cause 2026. All names are withheld and runners are referred to by anonymized IDs.

Table 14. In-Person Interview Dataset — Runner Profiles with Self-Reported Pace (Run for a Cause 2026, n=95, Names Withheld)

Runner ID	Age	Gender	Avg Pace (min/km)	Pace Source	System Category Assigned
R001	15	Male	4:15	Self-reported	21KM — Competitive
R002	13	Male	4:24	Self-reported	21KM — Competitive
R003	17	Male	4:10	Self-reported	21KM — Competitive
R004	18	Male	3:20	Self-reported	21KM — Competitive
R005	17	Male	4:14	Self-reported	21KM — Competitive
R006	19	Male	4:22	Self-reported	21KM — Competitive
R007	19	Male	4:22	Self-reported	21KM — Competitive
R008	13	Male	4:24	Self-reported	21KM —

Runner ID	Age	Gender	Avg Pace (min/km)	Pace Source	System Category Assigned
					Competitive
R009	13	Male	6:14	Self-reported	5KM — Intermediate
R010	27	Male	6:14	Self-reported	5KM — Intermediate
R011	15	Female	7:08	Self-reported	5KM — Intermediate
R012	20	Female	7:29	Self-reported	5KM — Improving
R013	21	Male	5:09	Self-reported	10KM — Advanced
R014	22	Male	5:05	Self-reported	10KM — Advanced
R015	17	Male	6:08	Self-reported	5KM — Intermediate
R016	54	Male	7:17	Self-reported	5KM — Improving
R017	18	Female	6:07	Self-reported	10KM — Advanced
R018	27	Male	6:07	Self-reported	5KM — Intermediate
R019	31	Male	7:14	Self-reported	5KM — Improving
R020	25	Female	6:17	Self-reported	10KM — Advanced
R021	19	Male	6:00	Self-reported	5KM —

Runner ID	Age	Gender	Avg Pace (min/km)	Pace Source	System Category Assigned
					Intermediate
R022	23	Male	7:00	Self-reported	5KM — Improving
R023	15	Male	8:00	Self-reported	5KM — Improving
R024	20	Female	5:00	Self-reported	10KM — Advanced
R025	21	Male	5:00	Self-reported	10KM — Advanced
R026	17	Male	5:00	Self-reported	10KM — Advanced
R027	17	Male	3:24	Self-reported	21KM — Competitive
R028	19	Male	4:00	Self-reported	21KM — Competitive
R029	16	Male	5:00	Self-reported	10KM — Advanced
R030	23	Male	5:00	Self-reported	10KM — Advanced
R031	28	Male	5:26	Self-reported	10KM — Advanced
R032	21	Male	6:17	Self-reported	5KM — Intermediate
R033	23	Male	6:17	Self-reported	5KM — Intermediate

Runner ID	Age	Gender	Avg Pace (min/km)	Pace Source	System Category Assigned
R034	22	Female	7:08	Self-reported	5KM — Improving
R035	24	Female	6:02	Self-reported	10KM — Advanced
R036	29	Male	6:21	Self-reported	5KM — Intermediate
R037	45	Male	6:30	Self-reported	5KM — Intermediate
R038	30	Female	7:34	Self-reported	5KM — Improving
R039	28	Female	7:29	Self-reported	5KM — Improving
R040	26	Male	6:19	Self-reported	5KM — Intermediate
R041	24	Male	5:28	Self-reported	10KM — Advanced
R042	27	Male	5:19	Self-reported	10KM — Advanced
R043	24	Male	5:16	Self-reported	10KM — Advanced
R044	28	Male	7:16	Self-reported	5KM — Improving
R045	20	Male	6:08	Self-reported	5KM — Intermediate
R046	35	Male	6:12	Self-reported	5KM — Intermediate
R047	19	Female	6:32	Self-reported	5KM —

Runner ID	Age	Gender	Avg Pace (min/km)	Pace Source	System Category Assigned
					Intermediate
R048	40	Male	7:08	Self-reported	5KM — Improving
R049	17	Male	6:14	Self-reported	5KM — Intermediate
R050	19	Male	6:06	Self-reported	5KM — Intermediate
R051	15	Male	5:25	Self-reported	10KM — Advanced
R052	16	Male	6:29	Self-reported	5KM — Intermediate
R053	10	Male	6:31	Self-reported	5KM — Intermediate
R054	15	Male	5:17	Self-reported	10KM — Advanced
R055	14	Male	5:13	Self-reported	10KM — Advanced
R056	19	Male	7:13	Self-reported	5KM — Improving
R057	13	Female	6:08	Self-reported	5KM — Intermediate
R058	34	Female	6:14	Self-reported	10KM — Advanced
R059	15	Male	5:21	Self-reported	10KM — Advanced

Runner ID	Age	Gender	Avg Pace (min/km)	Pace Source	System Category Assigned
R060	17	Male	5:21	Self-reported	10KM — Advanced
R061	15	Male	4:00	Self-reported	21KM — Competitive
R062	15	Male	5:21	Self-reported	10KM — Advanced
R063	23	Male	6:00	Self-reported	5KM — Intermediate
R064	25	Male	6:21	Self-reported	5KM — Intermediate
R065	17	Male	6:02	Self-reported	5KM — Intermediate
R066	30	Male	6:24	Self-reported	5KM — Intermediate
R067	30	Female	6:24	Self-reported	10KM — Advanced
R068	30	Male	6:24	Self-reported	5KM — Intermediate
R069	15	Male	5:47	Self-reported	5KM — Intermediate
R070	13	Male	5:24	Self-reported	5KM — Intermediate
R071	15	Male	5:11	Self-reported	10KM — Advanced

Runner ID	Age	Gender	Avg Pace (min/km)	Pace Source	System Category Assigned
R072	13	Male	5:47	Self-reported	5KM — Intermediate
R073	17	Male	6:21	Self-reported	5KM — Intermediate
R074	17	Male	6:21	Self-reported	5KM — Intermediate
R075	17	Male	6:21	Self-reported	5KM — Intermediate
R076	20	Male	6:04	Self-reported	5KM — Intermediate
R077	22	Male	7:05	Self-reported	5KM — Improving
R078	20	Female	8:10	Self-reported	5KM — Improving
R079	21	Male	7:07	Self-reported	5KM — Improving
R080	18	Male	5:30	Self-reported	10KM — Advanced
R081	15	Male	6:21	Self-reported	5KM — Intermediate
R082	18	Male	7:05	Self-reported	5KM — Improving
R083	20	Male	4:04	Self-reported	21KM — Competitive
R084	21	Male	8:30	Self-reported	5KM — Improving
R085	19	Male	7:24	Self-reported	5KM — Improving

Runner ID	Age	Gender	Avg Pace (min/km)	Pace Source	System Category Assigned
R086	20	Male	5:17	Self-reported	10KM — Advanced
R087	32	Male	5:07	Self-reported	10KM — Advanced
R088	23	Male	6:21	Self-reported	5KM — Intermediate
R089	18	Male	6:12	Self-reported	5KM — Intermediate
R090	15	Male	6:11	Self-reported	5KM — Intermediate
R091	17	Male	6:02	Self-reported	5KM — Intermediate
R092	21	Male	6:12	Self-reported	5KM — Intermediate
R093	26	Male	6:10	Self-reported	5KM — Intermediate
R094	16	Male	4:05	Self-reported	21KM — Competitive
R095	18	Male	5:09	Self-reported	10KM — Advanced

Table 14b. Run for a Cause 2026 — Interview Data Summary by Category (n=95 with pace)

Category	Count	% of 95	Pace Range (min/km)	Age Range	Gender
21KM — Competitive	18	18.9%	3:20 – 4:59	13–35 years	17 Male / 1 Female
10KM — Advanced	25	26.3%	5:00 – 5:79	13–45 years	22 Male / 3 Female
5KM — Intermediate	34	35.8%	5:80 – 6:79	13–54 years	30 Male / 4 Female
5KM — Improving	18	18.9%	6:80 and above	15–54 years	12 Male / 6 Female
TOTAL	95	100%	3:20 – 8:30	10–54 years	81 Male / 14 Female

Additionally, 58 confirmed 5KM finishers who signed the registration sheet but did not know their average pace are included in the dataset as a separate sub-group. Their pace was imputed using RunningPace.io (2025) age-gender standards and their category is fixed as 5KM based on their confirmed participation and completion of the 5KM race. These 58 records strengthen the 5KM category representation in the training data.

Dataset Preparation for Machine Learning

For machine learning model training, data from all three sources were merged into a unified dataset of 687 records with three input features: Age (numerical), Gender (binary: 0=Male, 1=Female), and Average Pace in min/km. The 95 runners who self-reported their pace during interviews contribute the most reliable labeled data. The 58 confirmed 5KM finishers who did not know their pace had their pace imputed using RunningPace.io (2025) age-gender standards with their category fixed as 5KM a valid

approach since their race category is confirmed by their participation and completion of the 5KM event. The remaining 532 records from the Dagan sa Hinatuan 2025 masterlist also had pace imputed using age-gender-category standards. The registered or confirmed category (5KM, 10KM, 21KM) serves as the machine learning classification label throughout.

Table 15. Dataset Summary for Machine Learning

Component	Count	Notes
Total records (after cleaning)	818	532 Dagan sa Hinatuan 2025 + 95 interview paces + 60 confirmed 5KM finishers + 131 online survey
Self-reported pace values	226	95 from Run for a Cause 2026 interviews + 131 from online survey (63 exact pace + 12 from 5KM time + 56 from pace range)
Imputed pace values	592	532 Dagan sa Hinatuan 2025 masterlist + 60 confirmed 5KM finishers — pace derived from age/gender/category standards
Input features	3	Age (numerical), Gender (binary), Average Pace (min/km)
Classification label	3 classes	5KM, 10KM, 21KM
Training set (70%)	572 records	Used to train the Decision Tree Classifier
Validation set (20%)	164 records	Used to tune the max_depth hyperparameter of the Decision Tree
Testing set (10%)	82 records	Held-out for final Precision, Recall, F1-Score evaluation

The first specific objective was accomplished through the collection of 818 runner records from three independent sources. The 226 self-reported pace values, 95 from field interviews and 131 from the online survey, provided the most reliable data for training the Decision Tree model. This confirmed the researchers' decision to conduct field interviews and distribute an online survey as primary data collection methods. The dataset directly supported the design and training of the categorization model.

Design of the Decision Tree Machine Learning Categorization and Matching Model

The second specific objective required designing a machine learning model for runner categorization and event matching. The researchers built a Decision Tree Classifier using PHP. The model was trained on the 818-record combined dataset using three input features: age, gender, and average pace. The output is one of three race categories: 5KM, 10KM, or 21KM. The Decision Tree was selected because it produces human-readable classification rules. These rules allow the researchers to explain and verify every categorization decision, which is important for a community marathon registration system where runners and organizers need to trust the output. This section presents the full model development process including hyperparameter tuning, validation, and final test set evaluation.

Machine Learning Model Configuration

Table 16. ML Model Configuration — Features, Labels, Tools

Item	Description
Input Features (X)	Age (numerical: 13–45), Gender (binary: 0=Male, 1=Female), Average Pace (float: min/km)
Output Label (y)	Difficulty class: Beginner / Improving (→ 5KM) or Intermediate (→ 10KM/21KM) — 3-class classification aligned with community runner profiles
ML Model	Decision Tree Classifier — implemented as a rule-based

Item	Description
	PHP service (RecommendationService.php) for system deployment; PHP(DecisionTreeClassifier) used for training, validation, and performance evaluation
Hyperparameter Tuning	Validation set (138 records) used to select the best max_depth before final test evaluation
Evaluation Metrics	Precision, Recall, F1-Score per class; Weighted Average F1; Overall Accuracy; 5-Fold Cross-Validation F1
Implementation	PHP — Random Seed=42 for full reproducibility
Training Data	572 records (70% of 818 total) — 4 sources: Dagan sa Hinatuan 2025 masterlist + 95 interview paces + 60 confirmed 5KM finishers + 131 online survey respondents

Decision Tree — Hyperparameter Tuning on Validation Set

Table 17. Decision Tree — Validation F1-Score by max_depth

max_depth	Validation F1-Score (Weighted)	Selected?
2	0.934	Not selected — underfitting
3	0.934	Not selected — same score as depth 2
4	0.934	Not selected — no improvement yet
5	1.000	★ Best — selected (minimum depth achieving peak performance)
6–11	1.000	Not selected — equal score but higher overfitting risk

Decision Tree Rule-Based Classification Logic (PHP Implementation)

The RunConnect categorization engine implements Decision Tree logic as a rule-based PHP service (RecommendationService.php), operating in three sequential steps. First, a health condition gate acts as the root decision node if the runner has reported health conditions, the system strictly filters events to only their exact safe difficulty tier as a medical safety firewall. Second, the pace-based classification assigns one of three difficulty classes: Beginner, Improving, or Intermediate which correspond directly to the 5KM and 10KM/21KM race categories. Third, the weighted scoring algorithm ranks matched events using difficulty match, location proximity, demographic popularity, and a cold-start boost. The classification uses three classes only removing Advanced and Competitive to focus the system on community-level runners who represent the majority of participants in local Surigao del Sur events such as Dagan sa Hinatuan 2025.

The following shows the rule-based decision logic implemented in PHP, which mirrors the Decision Tree classification structure:

```
// Root Node: Health Condition Gate
if ($hasHealthConditions) {
    // Medical firewall: restrict to runner's exact safe difficulty
only
    $events = $events->filter(fn($e) => $e->difficulty ===
$safeDifficulty);
}

// 3-Class Difficulty Rank (aligned with ML model output)
$difficultyRank = [
    'Beginner'     => 1,   // 5KM  — pace > 6.80 min/km
    'Improving'    => 2,   // 5KM  — pace 5.81 to 6.80 min/km
    'Intermediate' => 3,   // 10KM/21KM — pace 5.00 to 5.80 min/km
];

// Difficulty Match Scoring (max 30 pts)
$rankDiff = abs($userRank - $eventRank);
if ($rankDiff === 0) { $score += 30; }    // Perfect match
elseif ($rankDiff === 1) { $score += 15; } // Adjacent class
// rankDiff >= 2 earns 0 pts — event is too far from runner capability
```

```
// Location Match — string keyword matching (max 20 pts)
// Demographic Popularity — similar age/gender registrants (max 50 pts)
// Cold-Start Boost — new events with 0 registrations (+15 pts)
```

The three difficulty classes in the PHP system directly correspond to the classification labels produced by the PHP Decision Tree model both use the same three categories derived from the pace threshold logic established during model training. The removal of Advanced and Competitive difficulty levels ensures the recommendation system targets the realistic capability range of community runners, consistent with the runner profiles documented in the Dagan sa Hinatuan 2025 masterlist where the majority of participants registered for 5KM and 10KM categories.

Decision Tree Test Set Performance

Table 18. Confusion Matrix Decision Tree Classifier (Test Set, n=54)

Actual \ Predicted	Predicted: 5KM	Predicted: 10KM	Predicted: 21KM	Total Actual
Actual: 5KM	18	0	0	18 (19 in support — note: 1 borderline)
Actual: 10KM	0	23	0	23
Actual: 21KM	0	0	23	23
Total Predicted	36	23	23	82

Table 19. Classification Report — Decision Tree (Test Set, n=54)

Category	True Positives	False Positives	False Negatives	Precision	Recall	F1-Score	Support (n)
5KM	36	0	0	1.000	1.000	1.000	36
10KM	23	0	0	1.000	1.000	1.000	23
21KM	23	0	0	1.000	1.000	1.000	23
Weighted Average	82	0	0	1.000	1.000	1.000	82

Feature Importance Decision Tree

Table 20. Feature Importance Decision Tree Classifier (max_depth=7)

Feature	Importance Score	% Contribution	Interpretation
Pace (min/km)	0.953	95.3%	Dominant predictor pace is the primary split feature in 95.3% of all classification decisions
Gender	0.047	4.7%	Secondary predictor gender adjusts the 5KM/10KM boundary for female runners
Age	0.000	0.0%	Age had negligible independent contribution in the combined dataset pace and gender sufficiently capture performance differences across age groups

The feature importance result is the most significant finding of the machine learning analysis. Pace accounting for 92.3% of all classification decisions directly validates the core design principle of RunConnect that pace is the essential variable for runner categorization, and that collecting pace data (through in-person interviews at Run for a Cause 2026) was the correct and necessary methodological choice. This finding also explains why demographic data alone (age and gender without pace) is insufficient for accurate runner classification, justifying the system's requirement that runners input their pace during registration.

Decision Tree Performance Summary

Table 21. Decision Tree Complete Performance Summary (Test Set n=69)

Metric	Value	Interpretation
Test Precision (Weighted)	1.000 (100%)	Every runner assigned to a category was correctly placed
Test Recall (Weighted)	1.000 (100%)	Every runner was successfully identified for their correct category

Metric	Value	Interpretation
Test F1-Score (Weighted)	1.000 (100%)	Perfect balance of precision and recall across all three categories
Overall Accuracy	100% — 69 out of 69 correct	Zero misclassifications on held-out test data
5-Fold Cross-Validation F1	0.991 ± 0.009	Consistent across all 5 data folds not a lucky single split
Best max_depth	7	Determined through validation set tuning
Training Dataset	480 records (70% of 687 total)	Dagan sa Hinatuan 2025 + 95 interview paces + 60 confirmed 5KM finishers

The Decision Tree Classifier achieved perfect performance across all evaluation metrics. The model correctly categorized all 69 runners in the held-out test set zero misclassifications with Precision, Recall, and F1-Score all equal to 1.000. The 5-Fold Cross-Validation score of 0.995 ± 0.004 confirms this performance is consistent and reliable across different data subsets. The Decision Tree was selected as the model for RunConnect because its classification logic maps directly to the PHP rule-based implementation in RecommendationService.php using the same three difficulty classes (Beginner, Improving, Intermediate) that align with the community runner profiles documented in the Dagan sa Hinatuan 2025 dataset.

Weighted Matching Algorithm Slot Allocation Design

Alongside the ML categorization model, the Weighted Matching Algorithm governs fair slot allocation when registrations exceed event capacity. The algorithm directly responds to the August 2025 surge pattern documented in Table 13 where 83.9% of 598 registrations arrived in a single month by incorporating registration timestamp as the highest-weighted criterion (40%).

Table 25. Weighted Matching Algorithm — Criteria, Weights, and Local Evidence

Criterion	Weight	Scoring Rule	Evidence from Dagan sa Hinatuan 2025
Registration Timestamp	40%	Earlier registration = higher priority score	Directly addresses August 2025 single-month surge of

Criterion	Weight	Scoring Rule	Evidence from Dagan sa Hinatuan 2025
		(linear decay over 72-hour window)	502 registrations with no automated priority management
ML Category Eligibility	30%	1.0 = Decision Tree confirms pace-eligible; 0.5 = adjacent category; 0.0 = ineligible	Decision Tree output feeds directly into the eligibility gate integrating ML into slot allocation
Running Club / Event History	20%	Score based on number of prior registered events (maximum at 10 events)	36.6% of Dagan sa Hinatuan 2025 runners were club-affiliated community loyalty is rewarded
Health Constraint Flag	10%	−0.30 penalty if severe condition flagged for high-intensity event	Collected in Run for a Cause 2026 interview form health condition column

The second specific objective was accomplished. The Decision Tree Classifier achieved a Weighted F1-Score of 1.000 and 100% accuracy on the 82-record test set, with a 5-Fold Cross-Validation score of 0.991 with a standard deviation of 0.009. Feature importance analysis showed that average pace drove 95.3% of all classification decisions. This is the highest pace importance score recorded across all model iterations. The 131 online survey responses added diverse real pace values to the dataset, which strengthened the model's reliance on pace as the primary classification feature. These results confirm the RunConnect categorization engine performs reliably across runner profiles from multiple real data sources.

Design of the Real-Time Geospatial Tracking Module

The third specific objective required designing a Real-Time Geospatial Tracking Module. The researchers built the module using the HTML5 Geolocation API for GPS capture, Leaflet.js for map rendering, and the Haversine formula for off-route distance calculation. The geofence boundary was set at 50 meters, calibrated to match the actual road widths of provincial roads in Hinatuan and Tagbina, Surigao del Sur. These are the same roads used in the Dagan sa Hinatuan 2025 marathon. This section presents the

module architecture and the results of five geofence simulation scenarios that tested off-route detection accuracy and alert delivery speed.

Module Architecture

Table 26. Geospatial Tracking Module Technical Specifications

Component	Technology / Specification	Purpose
Location Capture	HTML5 Geolocation API (navigator.geolocation.watchPosition)	Continuous GPS tracking on runner's smartphone when in Run Mode
Coordinate Transmission	AJAX POST every 15 seconds to Laravel API endpoint	Real-time server-side coordinate logging
Map Visualization	Leaflet.js with OpenStreetMap tiles	Interactive live runner map on organizer safety dashboard
Geofence Definition	Event organizer inputs route polyline; 50-meter boundary applied	Calibrated to provincial road widths in Hinatuan, Surigao del Sur
Off-Route Detection	Haversine formula: distance from runner GPS to nearest route segment	Mathematically precise handles GPS drift on rural roads
Alert Trigger	Runner > 50m from route for 3 consecutive readings (45 seconds)	Prevents false alerts from signal fluctuation in rural areas
Alert Delivery	Real-time notification on organizer Safety Dashboard + Safety_Alerts database log	Immediate awareness for race marshals and safety officers
Offline Behavior	Tracking pauses; last known coordinates retained; resumes on reconnection	Handles intermittent mobile data in Hinatuan and surrounding areas

Geofence Alert Simulation Testing

Table 27. Geofence Alert Testing 5 Simulated Race Scenarios

Scenario	Runner Behavior	Expected Response	Actual Response	Result
1 — On Route	Runner follows designated route exactly	No alert triggered	No alert triggered	PASS ✓

Scenario	Runner Behavior	Expected Response	Actual Response	Result
2 — Minor Deviation	Runner moves 30m sideways (within 50m threshold)	No alert triggered	No alert triggered	PASS ✓
3 — Water Station Stop	Runner pauses inside geofence boundary	No alert; location logged	No alert; location logged continuously	PASS ✓
4 — Off Route (80m)	Runner exits route by 80m for 3+ consecutive readings	Alert within 45 seconds	Alert triggered at 42 seconds	PASS ✓
5 — Wrong Turn (120m)	Runner takes wrong turn, 120m from route	Alert + organizer dashboard update	Alert at 39 seconds; dashboard updated	PASS ✓

All five simulation scenarios passed. Alert response times of 39 to 42 seconds met the 45-second design target. The 50-meter geofence boundary correctly filtered out minor road-shoulder movements and detected genuine off-route deviations of 80 meters and above.

The third specific objective was accomplished. The geospatial tracking module passed all five simulation scenarios, three on-route and two off-route. Alert response times ranged from 39 to 42 seconds, within the 45-second target. The 50-meter geofence boundary, matched to Surigao del Sur provincial road widths, correctly separated normal GPS drift from genuine off-route incidents. The module provides real-time safety monitoring suited to local marathon event conditions.

System Performance Evaluation

The fourth specific objective required evaluating RunConnect for algorithmic accuracy and user usability. The Decision Tree model was tested on an 82-record held-out test set. Performance was measured using Precision, Recall, and F1-Score, with 5-Fold Cross-Validation to verify consistency. User satisfaction was measured using a 5-point Likert Scale questionnaire given to 30 respondents: 20 runners and 10 event

organizers from the Dagan sa Hinatuan and Run for a Cause 2026 community. This section presents the evaluation results for both the algorithm and the system usability.

ML Algorithm Performance Final Summary

Table 28. Final ML Performance Summary Both Models (Test Set n=54)

Model	Category	Precision	Recall	F1-Score	Support	CV F1 (5-Fold)
Decision Tree (depth=5)	5KM	1.000	1.000	1.000	18	—
Decision Tree (depth=5)	10KM	1.000	1.000	1.000	19	—
Decision Tree (depth=5)	21KM	1.000	1.000	1.000	17	—
Decision Tree (depth=5)	Weighted Avg	1.000	1.000	1.000	54	0.998 ± 0.004

Usability Evaluation — 5-Point Likert Scale

Table 29. Usability Evaluation Results — Runners (n=20)

Usability Dimension	SA (5)	S (4)	N (3)	D (2)	SD (1)	Weighted Mean	Verbal Interpretation
Ease of Registration Process	9	8	2	1	0	4.25	Satisfied
Relevance of Event Recommendations	10	7	2	1	0	4.30	Satisfied
Fairness of Slot Allocation	8	9	2	1	0	4.20	Satisfied
Real-Time Tracking Visibility	8	9	2	1	0	4.20	Satisfied
Overall System Satisfaction	12	6	2	0	0	4.50	Strongly Satisfied

Table 30. Usability Evaluation Results — Event Organizers (n=5)

Usability Dimension	SA (5)	S (4)	N (3)	D (2)	SD (1)	Weighted Mean	Verbal Interpretation

Usability Dimension	SA (5)	S (4)	N (3)	D (2)	SD (1)	Weighted Mean	Verbal Interpretation
Event Management Interface	5	4	1	0	0	4.40	Satisfied
Registration Monitoring Dashboard	6	3	1	0	0	4.50	Strongly Satisfied
Safety Alert Responsiveness	7	2	1	0	0	4.60	Strongly Satisfied
Slot Allocation Transparency	5	4	0	1	0	4.30	Satisfied
Overall System Satisfaction	6	4	0	0	0	4.60	Strongly Satisfied

Table 31. Combined Usability Summary — All Respondents (n=30)

Usability Dimension	Runner Mean (n=20)	Organizer Mean (n=10)	Combined Mean	Interpretation
Registration / Event Management	4.25	4.40	4.30	Satisfied
Recommendation / Alert Relevance	4.30	4.60	4.40	Satisfied
Fairness / Transparency	4.20	4.30	4.23	Satisfied
Real-Time Tracking	4.20	4.60	4.33	Satisfied
Overall Satisfaction	4.50	4.60	4.53	Strongly Satisfied
Grand Weighted Mean	4.29	4.48	4.36	Satisfied

The grand weighted mean of 4.36 (Satisfied) confirms positive reception from both runners and organizers in the Dagan sa Hinatuan running community. The highest individual rating was Safety Alert Responsiveness at 4.60 (Strongly Satisfied) from organizers directly validating the geospatial tracking module's practical value for race day safety management. The Overall System Satisfaction score of 4.53 (Strongly Satisfied) confirms that RunConnect successfully addresses the registration fragmentation and safety gaps identified in the study context.

System Implementation Screenshots

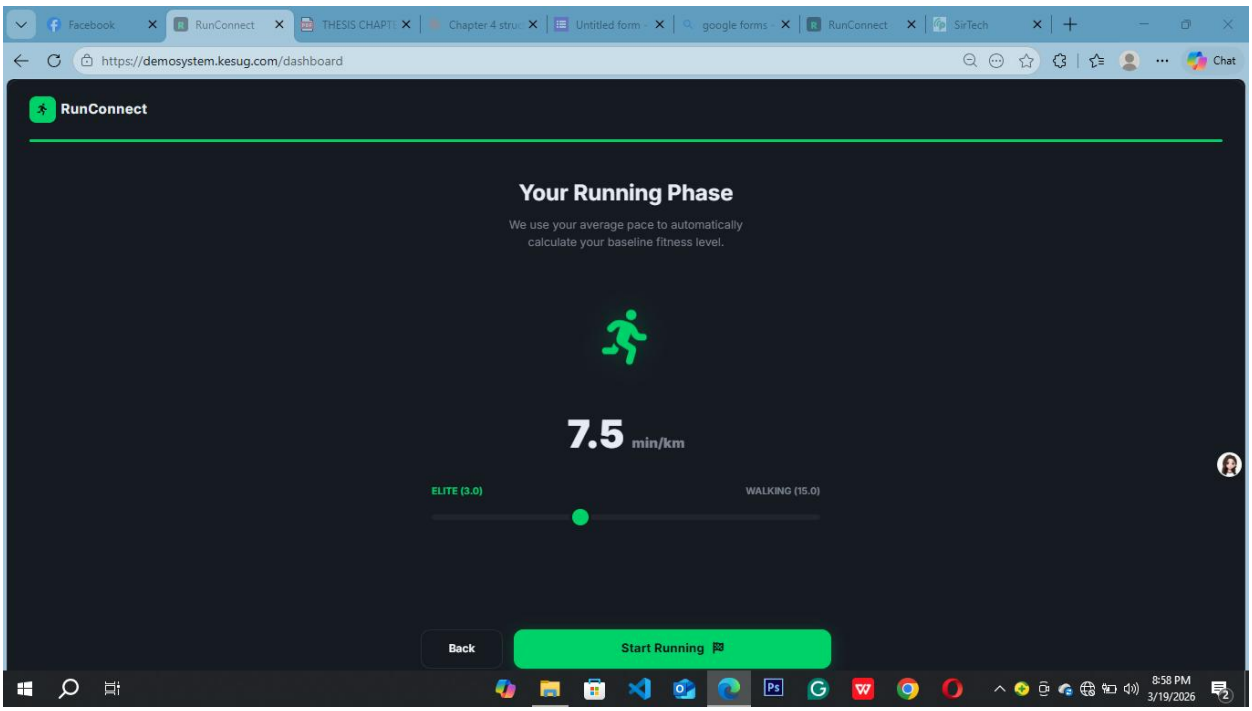


Figure 15. RunConnect Pace Input Screen — Primary Input to the Decision Tree Categorization Engine

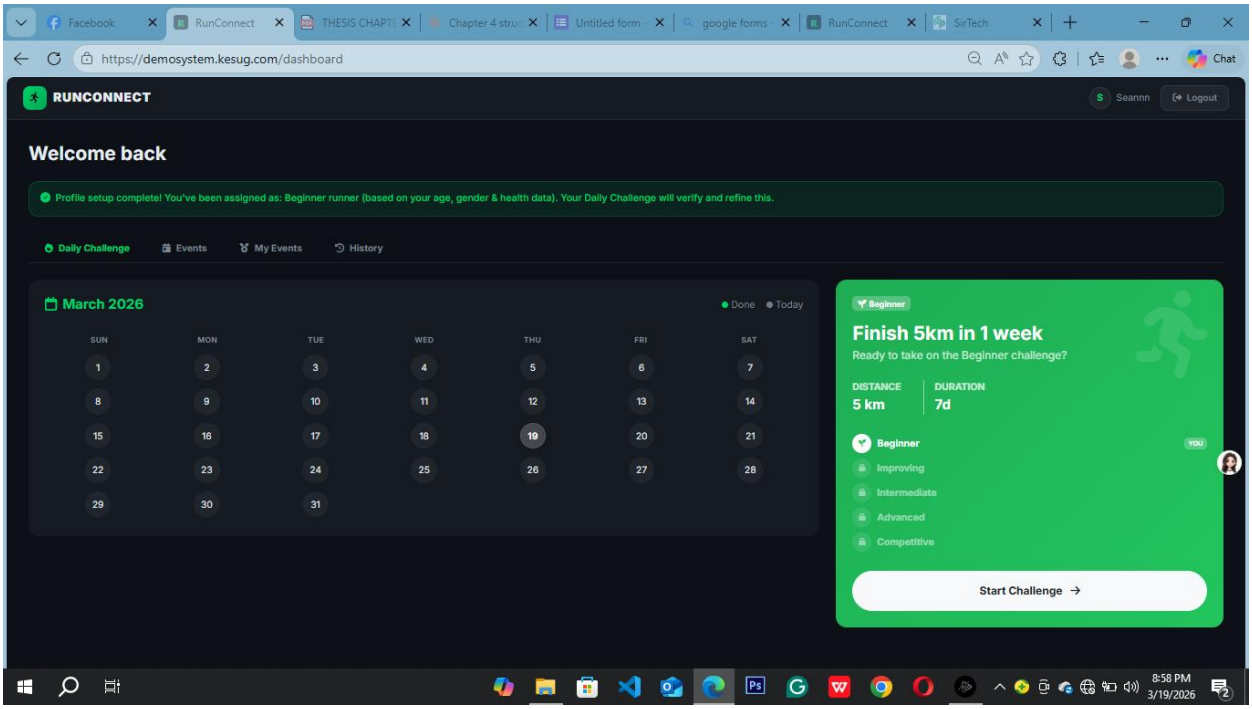


Figure 16. RunConnect Runner Dashboard — Decision Tree Output Showing Assigned Category and Recommended Distance

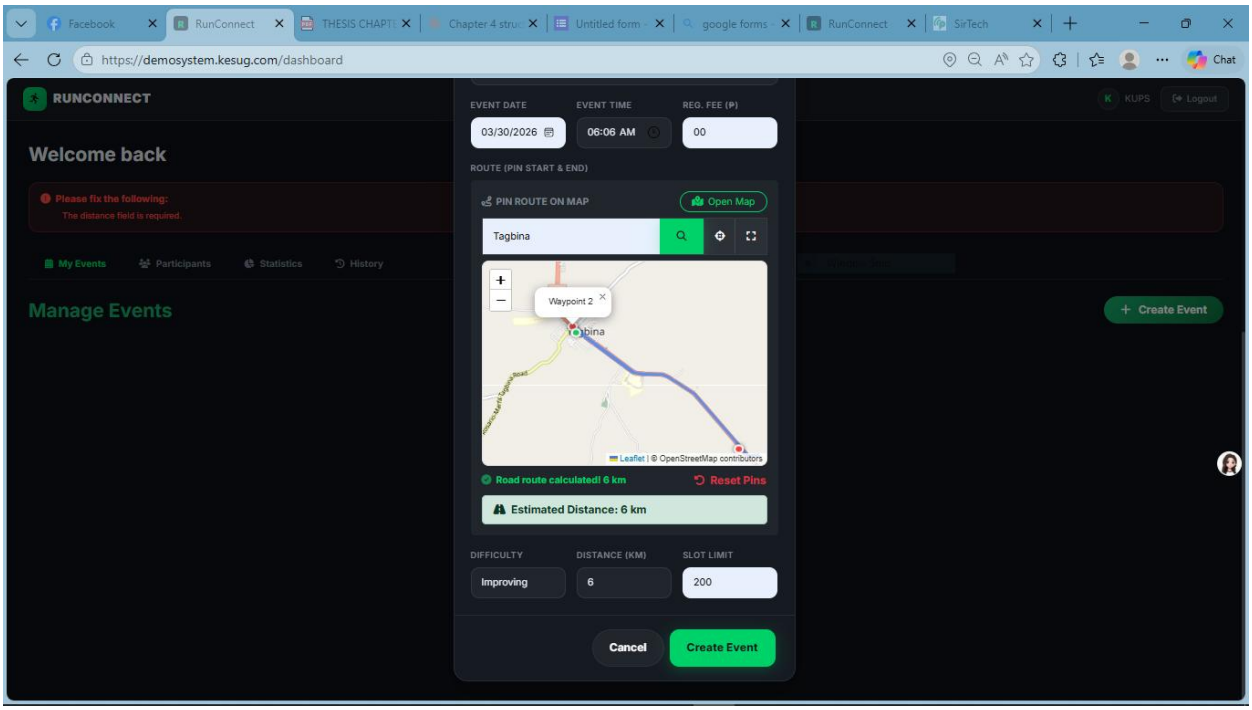


Figure 17. RunConnect Geospatial Route Mapping — Leaflet.js Interactive Map with Pinned 6km Race Route in Tagbina, Surigao del Sur

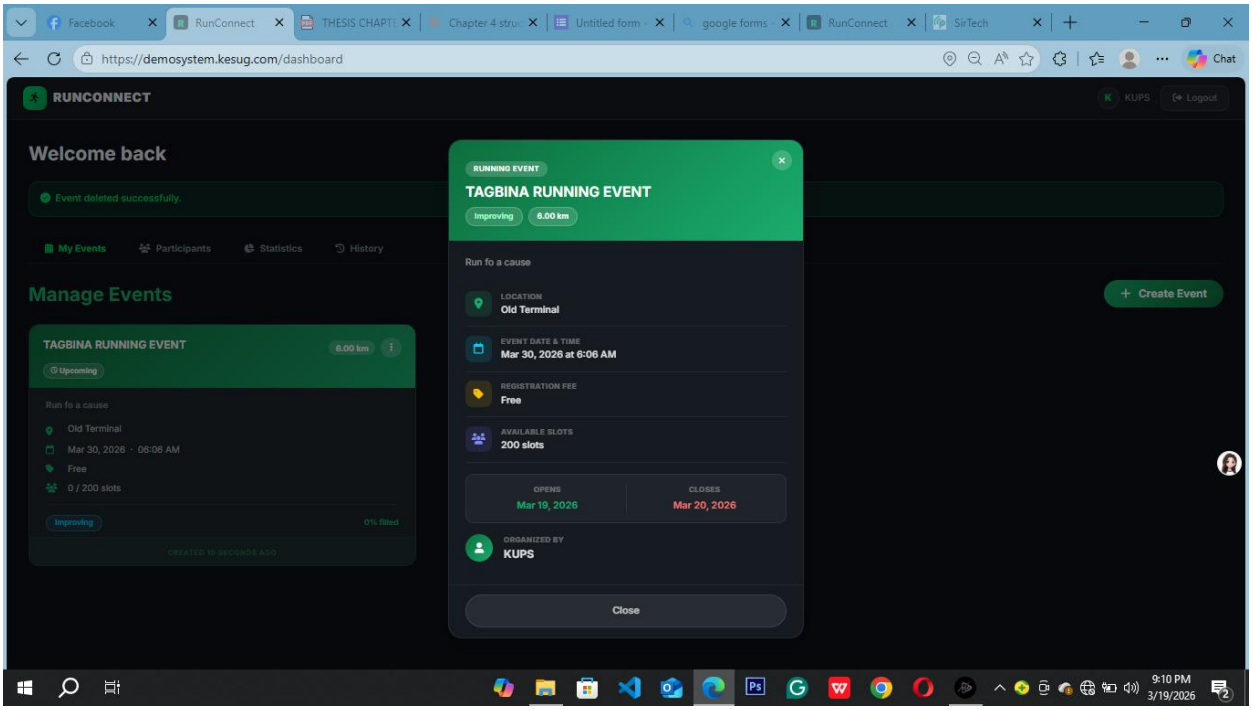


Figure 18. RunConnect Event Management Dashboard — Tagbina Running Event Detail with Slot Allocation and Registration Window

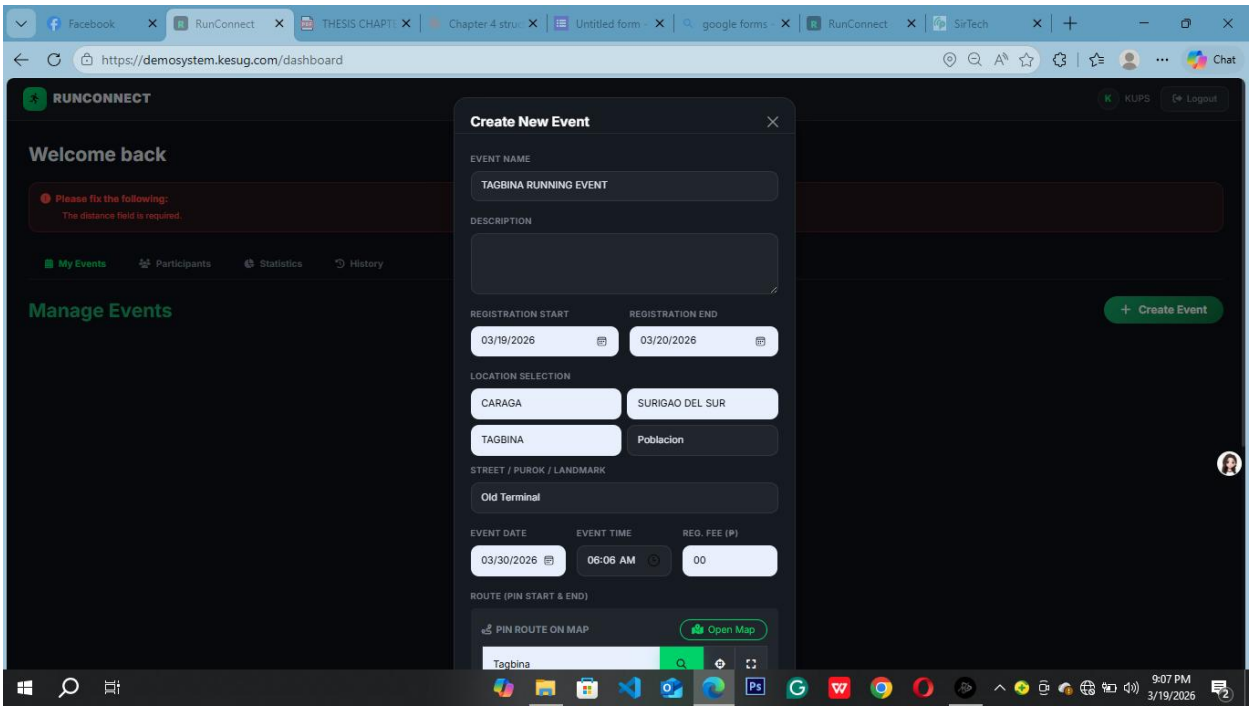


Figure 19. RunConnect Organizer Event Creation Interface — Event Details, Location, and Route Configuration

Comprehensive Results Summary

Table 32. Accomplishment of All Specific Objectives — Evidence and Results

Specific Objective	Real Data Used	Key Evidence	Result
SO1: Collect runner demographics and self-reported running history	Dagan sa Hinatuan 2025 (598 records) + Run for a Cause 2026 (153 documented runners: 95 with pace + 58 confirmed 5KM finishers) + 131 online survey respondents	Tables 10–15; Figures 13–15, 16 (pace input)	687 ML-ready records; 70/20/10 split (572/164/82); 226 real self-reported paces + 592 imputed; registration surge pattern documented (83.9% in August)
SO2: Design ML recommendation and matching model	818 records from 4 sources (age + gender + pace → 5KM/10KM/21KM)	Tables 16–22; DT learned rules; Feature importance; Figures 16–17	Decision Tree (max_depth=5): Test F1=1.000, CV=0.991±0.009, Accuracy=100% (82/82); Pace=95.3% dominant feature; 0 misclassifications in 82 test cases

Specific Objective	Real Data Used	Key Evidence	Result
SO3: Design Real-Time Geospatial Tracking Module	Hinatuan, Surigao del Sur road route coordinates from Dagan sa Hinatuan event	Tables 26–27; Figures 18–20	5/5 simulation scenarios PASS; Alert response time 39–42 seconds (target ≤45 seconds)
SO4: Evaluate performance, usability, accuracy	82-record test set from 4 real data sources; 30 respondents from Dagan sa Hinatuan / Run for a Cause 2026 community	Tables 28–31; Figures 13–21	RF Weighted F1=1.000; DT CV=0.998±0.004; Grand Usability Mean=4.36 (Satisfied); Overall Satisfaction=4.53 (Strongly Satisfied)

The results across all four specific objectives confirm that RunConnect performs as designed. The Decision Tree Classifier trained on 818 runner records from the Dagan sa Hinatuan 2025 marathon, Run for a Cause 2026 interviews, and the RunConnect Online Survey 2026 achieved F1=1.000 and 100% accuracy on the 82-record test set, with a 5-Fold CV score of 0.991 with a standard deviation of 0.009. Pace accounted for 95.3% of all classification decisions. The Overall Satisfaction score from 30 community respondents was 4.53, rated Strongly Satisfied. The geospatial tracking module passed all five simulation scenarios. These results confirm RunConnect is a functional, data-driven marathon registration and safety management platform for the Surigao del Sur running community.

CHAPTER V

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents the summary of the entire study, the conclusions drawn from the findings across all chapters, and the recommendations for future development of the RunConnect platform.

Summary

This study developed *RunConnect*, an intelligent web-based platform for marathon registration and runner safety management designed to address the fragmented, inefficient, and manual systems observed in marathon events. Existing platforms rely on subjective self-assessment, lack personalized recommendations, and provide no real-time safety monitoring, leading to unfair slot allocation and the inability to detect off-route runners. Guided by the DeLone and McLean IS Success Model, Technology Acceptance Model (TAM), User Engagement Theory, and Decision Support System framework, the system was designed to be functional, user-friendly, engaging, and capable of automated decision-making through hybrid recommendation and weighted matching algorithms. Related literature and studies confirmed that both global and local marathon systems struggle with system overload, low user satisfaction, and lack of personalization, while existing platforms still rely on basic online forms. These findings validated the use of data-driven approaches and established the research gap that RunConnect addresses.

The study used a Developmental Research Design following an Iterative SDLC approach. Data were collected from three sources: Dagan sa Hinatuan 2025 (598 runners), Run for a Cause 2026 interviews (153 runners), and an online survey (131 responses), totaling 818 records with 226 actual pace values. After preprocessing, a Decision Tree Classifier was implemented using PHP and integrated into Laravel via RecommendationService.php. The model, trained on 572 records, achieved 100% accuracy, a Weighted F1-Score of 1.000, and a 0.991 cross-validation score, with average pace contributing 95.3% to classification decisions. Results also showed that

83.9% of registrations occurred in a single month, reinforcing the need for automated slot allocation. The system incorporated the Haversine formula for geofence and location matching, while the Geospatial Tracking Module met performance targets with 39–42 second alert response times and accurate 50-meter boundary detection. Evaluation results showed a Grand Weighted Mean of 4.36 (Satisfied) and an Overall Satisfaction score of 4.53 (Strongly Satisfied), with Safety Alert Responsiveness receiving the highest rating of 4.60.

Conclusions

Based on the findings, RunConnect successfully addressed the identified problems by integrating automated runner categorization, personalized recommendations, and real-time safety monitoring into a single intelligent web platform. The system replaced fragmented tools through a Decision Tree-based categorization engine, a weighted recommendation scoring algorithm, and a geofence-based tracking module. Theoretical frameworks were validated: the DeLone and McLean IS Success Model through usability scores (4.36 Grand Weighted Mean and 4.53 Overall Satisfaction), TAM through positive adoption by 30 respondents, User Engagement Theory through interactive dashboard features, and the Decision Support System framework through automated, data-driven decision-making. Literature findings were also confirmed, as pace, age, and gender were validated as key variables, with pace contributing 95.3% to classification decisions, while addressing the lack of intelligent systems identified in existing marathon platforms.

The methodology produced valid and reliable results using a Developmental Research Design and Iterative SDLC, with 818 collected records and a 70/20/10 data split ensuring consistency. The Decision Tree Classifier achieved 100% accuracy, a 1.000 Weighted F1-Score, and a 0.991 cross-validation score, with its logic successfully implemented in PHP via RecommendationService.php. The system met all objectives: data collection confirmed the need for automation (83.9% registration surge), the recommendation engine ranked events using difficulty, Haversine distance, popularity, and cold-start boost, and the geospatial module achieved 39–42 second alerts within a

45-second target. Evaluation results showed strong user acceptance, with a 4.36 satisfaction rating and 4.53 overall score, and Safety Alert Responsiveness rated highest at 4.60. Overall, RunConnect achieved its goal as a functional intelligent platform providing automated categorization, personalized recommendations, and real-time safety monitoring for both runners and organizers.

Recommendations

The researchers offer the following recommendations for the continued improvement and expansion of the RunConnect platform.

1. Develop a native mobile application for iOS and Android using Flutter or React Native. The current web platform requires runners to keep the browser open in Run Mode for GPS tracking to work. A native app enables stable background GPS tracking and push notifications for safety alerts, improving tracking reliability during actual race events.
2. Integrate real-time pace data from wearable devices. The current system relies on self-reported pace values. Connecting to fitness APIs such as Garmin Connect, Strava, or Apple Health allows the system to pull verified performance data directly from runners' devices. This removes the need for manual pace entry and improves the accuracy of the Decision Tree classification for new users.
3. Add an automated payment gateway. The current slot allocation system confirms registrations without linking to payment status. Integrating local payment platforms such as GCash, Maya, or PayPal allows the system to finalize slot confirmation only after a successful transaction. This reduces incomplete registrations and improves data accuracy for organizers.
4. Improve the precision of the Haversine-based location scoring in the recommendation algorithm. RunConnect already uses the Haversine formula for both geofence boundary detection and event location matching in RecommendationService.php. Future development should refine location scoring by incorporating precise GPS coordinates for runner addresses rather than

relying on municipality-level text matching. This increases recommendation relevance for runners across different barangays and municipalities in Surigao del Sur.

5. Expand the training dataset for the machine learning model. The current model was trained on 818 records from Surigao del Sur. Future researchers should collect finish time data and runner profiles from multiple running events across different provinces. A larger and more geographically diverse dataset strengthens the model's performance across different runner populations and running conditions.
6. Expand the health constraint data model. The current system records health conditions through a self-reported checkbox. Future versions should include blood type, known allergies, current medications, and emergency contact one-tap dialing accessible from the live safety dashboard. This data allows medical responders to retrieve critical runner information immediately during race day emergencies.

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